



Roots and Shoots Foundation

Non-Profit Tree Conservation Organisation

User Centred Design

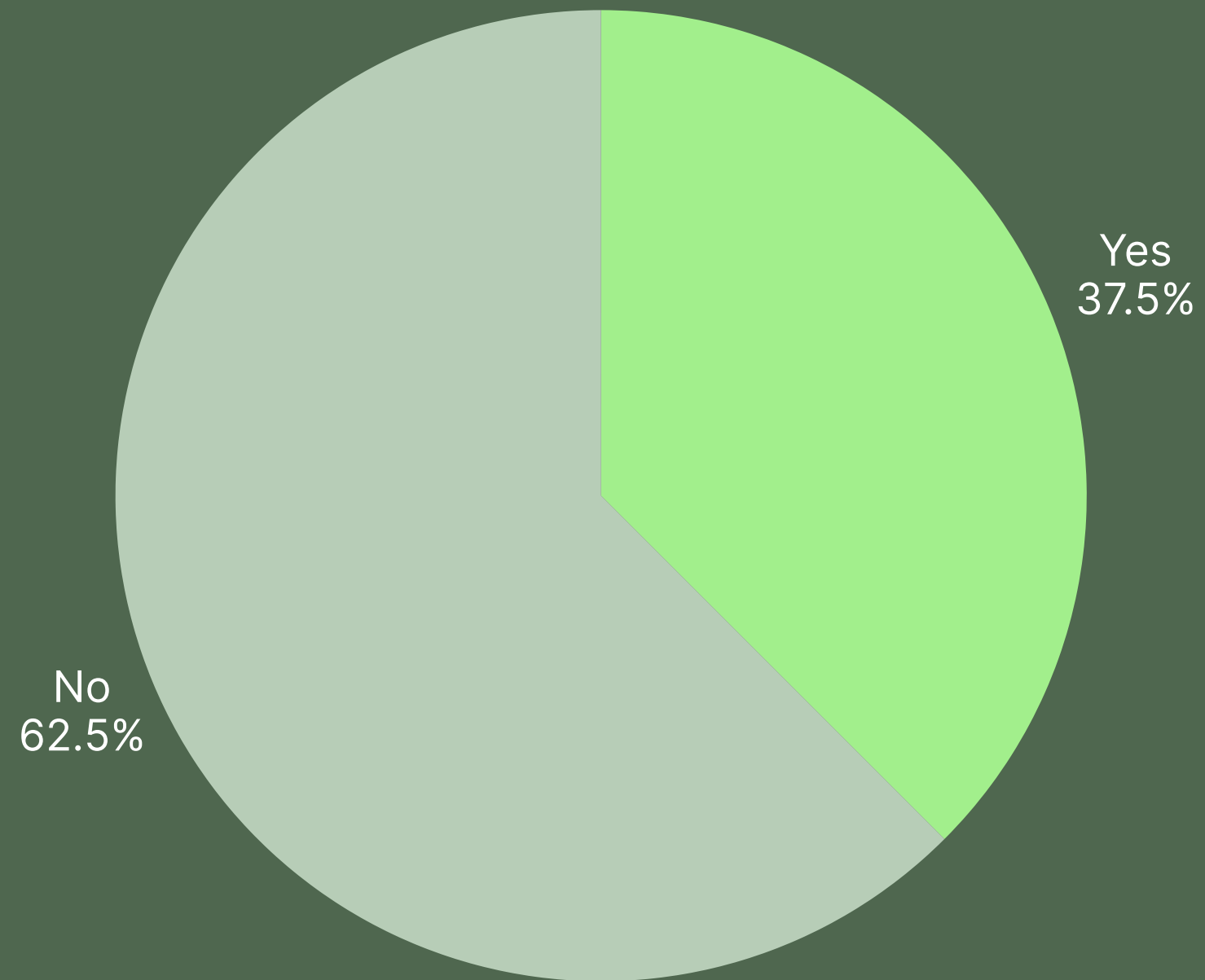


The Roots and Shoots Foundation website will utilise interactive features and gamification to present information regarding deforestation and the organisation's initiatives. The site will include colorful visuals, and live interactable content such as leaderboards, progress trackers and interactive maps to reflect real-time changes that will encourage user involvement and create a personal connection.

Product Description Statement

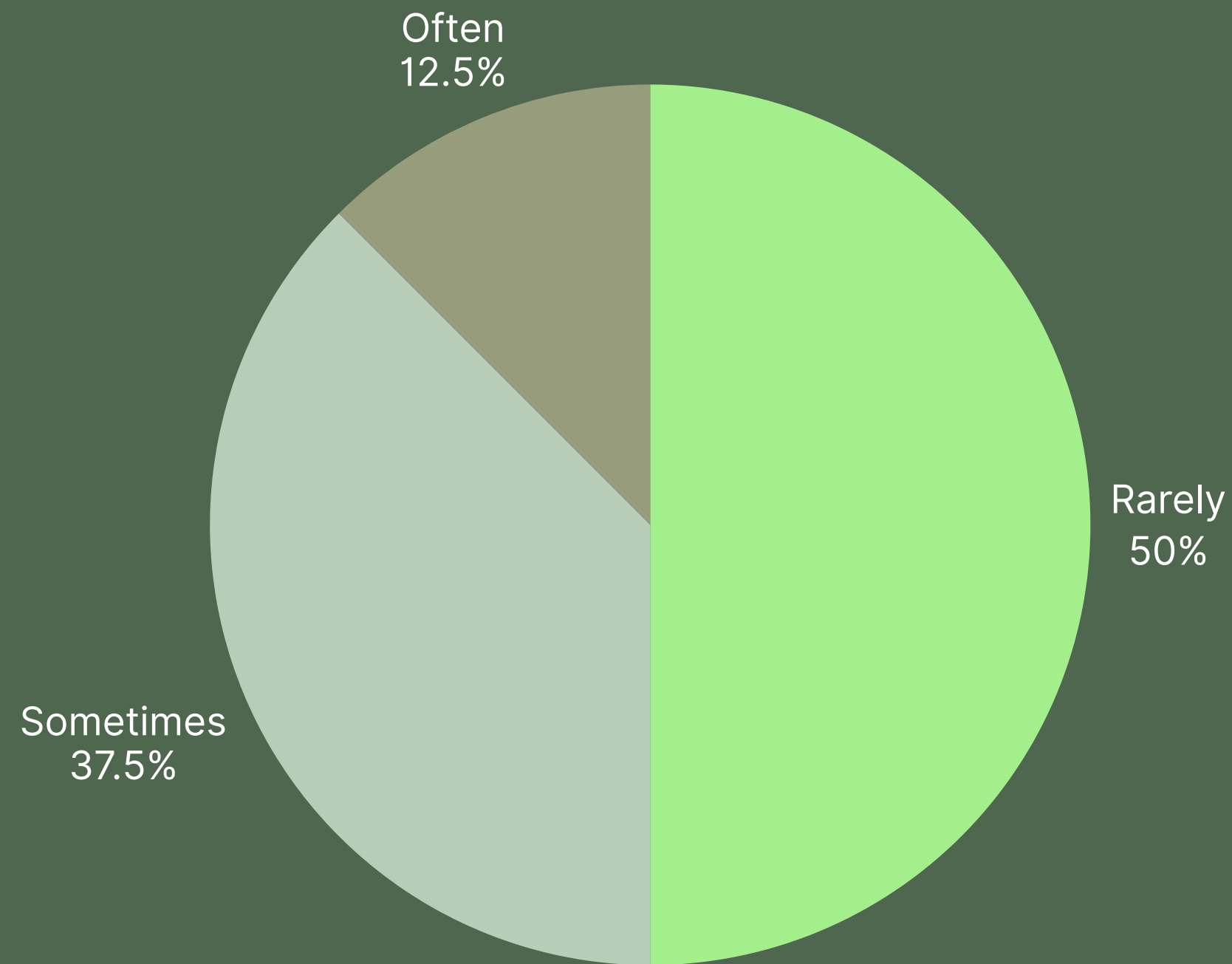
Interviews

Do you donate to charity?



Donation Tendency

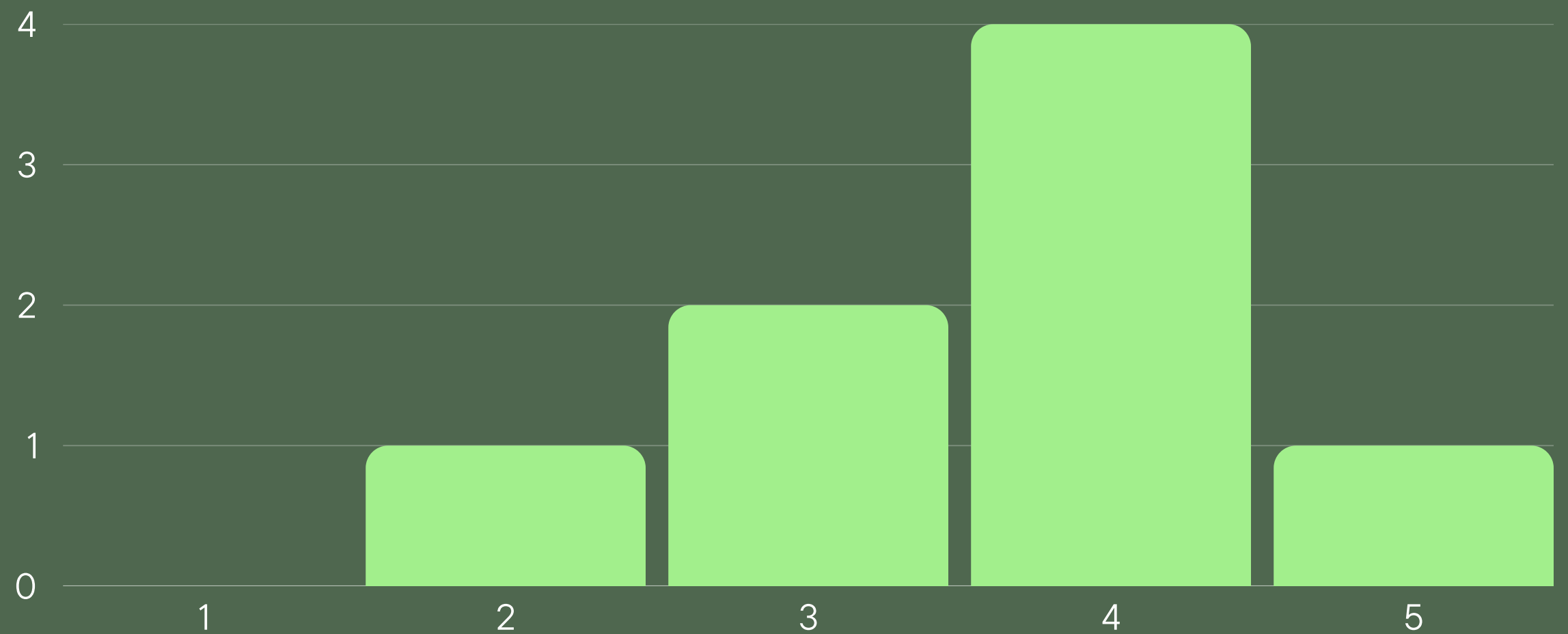
How frequently do you engage with content related to environmental issues online?



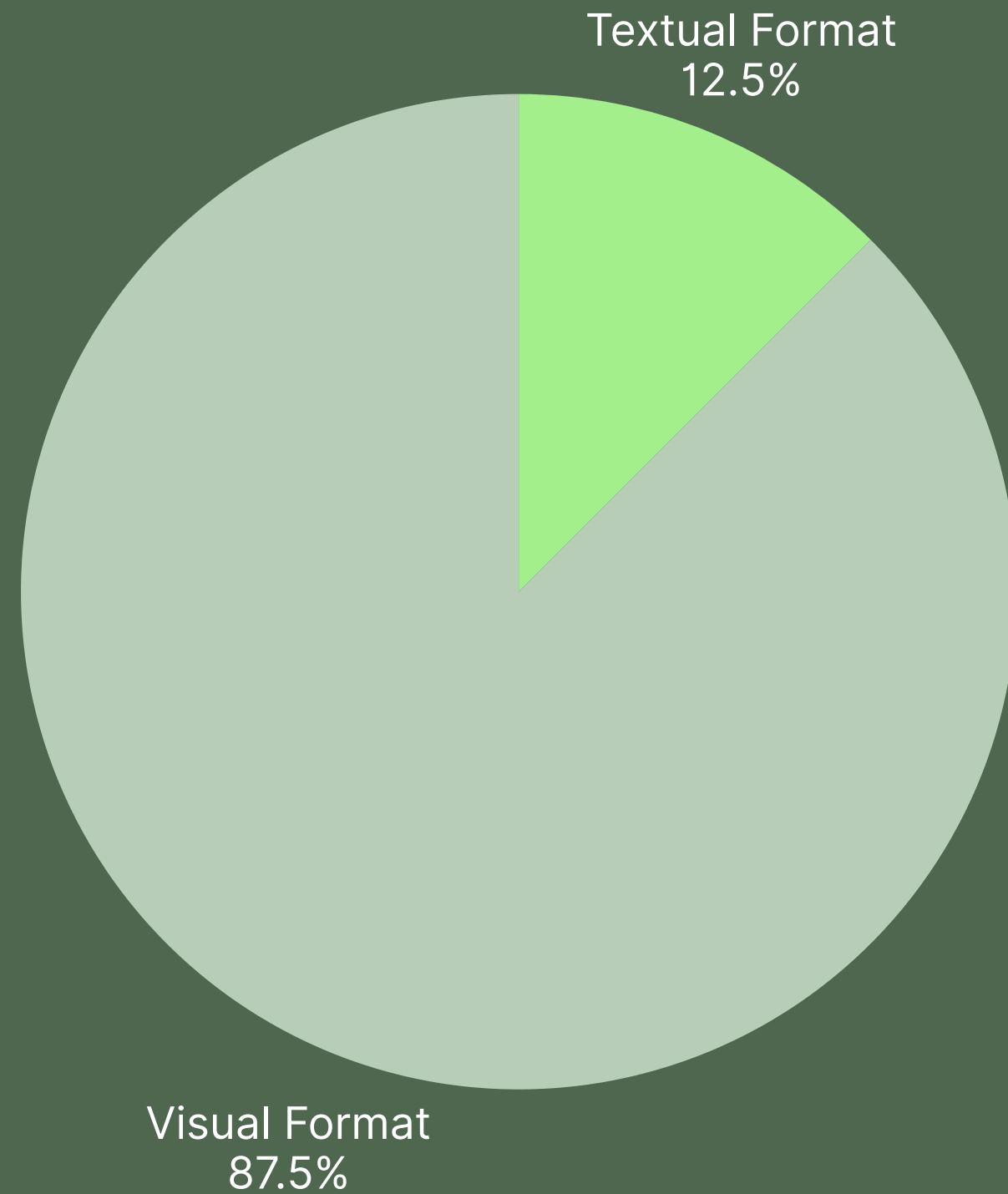
Online Content Interaction

3.63
Average Rating

How important is it for you to have an interactive website experience? (from 1 - 5)?

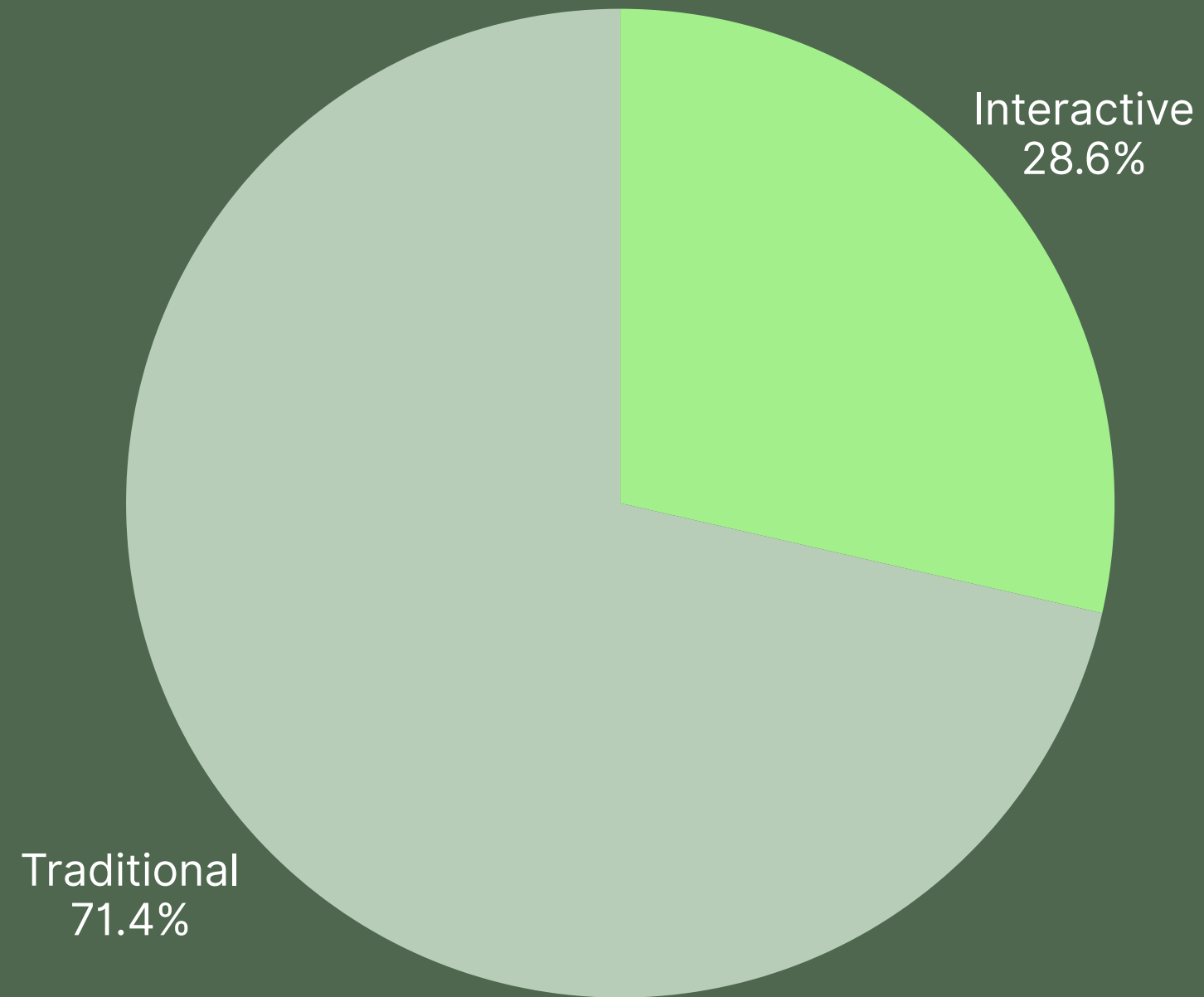


Do you prefer information to be presented in
a textual or visual format?



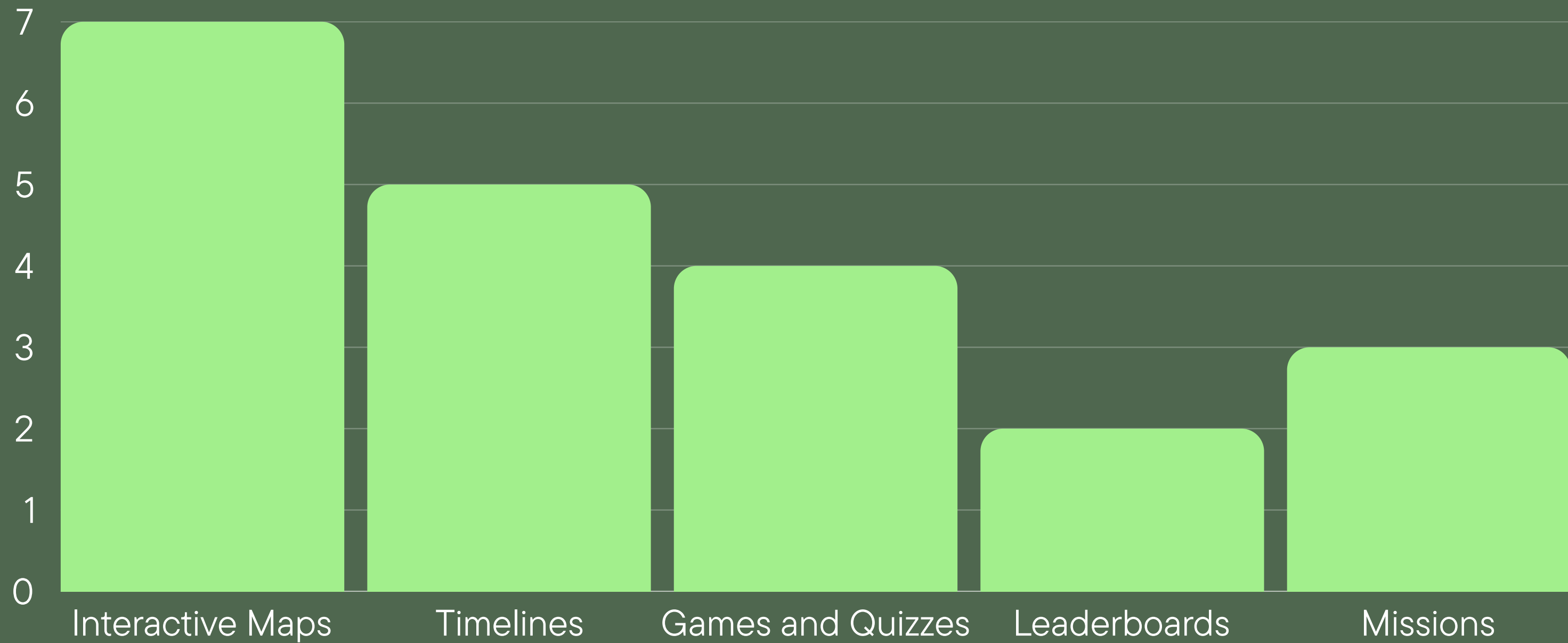
Format Preference

Would you be more interested in an Interactive,
Game-like website, or a Traditional website?



Website Structure

What interactive elements would you like to see on a website?



User Personas



Robin Harrell

CEO of a medium-sized firm
DevelopCorp.

Her goals include fulfilling
**corporate social
responsibility** to strengthen
DevelopCorp.'s **brand image**
by making a **tangible
positive impact** on the
environment



Ted Hastings

Retired Park Ranger with
early onset hearing loss
(presbycusis)

His goals include finding an
accessible and **simplistic**
way to stay up to date with
**current environmental
issues** given his **impairment**

User Personas



Raymond Shen

Current Undergraduate
student

His goals include **learning** about climate change and tree conservation in an **efficient** and **appealing** manner and **sharing** what he has learnt with **his friends**



Michaela Scott

Prominent local and online
Environmental Activist

Her goals include raising **public awareness** of deforestation through finding **community reforestation projects** to get people involved in

Context Scenarios



Robin Harrell

- Hears about The Roots and Shoots Foundation
- Appreciates the visual replanting statistics
- When viewing the leaderboard, she is impressed by the donations
- Robin completes the Foundation's Contact Form and reaches out via email



Ted Hastings

- Logs in with the intention to donate
- Finds article with accompanying video
- Video has transcript which helps with his hearing loss
- Article inspires him to donate
- Donation is a simple and seamless process

Context Scenarios



Raymond Shen

- Decides to do some research to inform himself
- Visits the website and is drawn in by the pleasing design and the uncluttered interface
- The resources allow him to learn without feeling pressured or overwhelmed

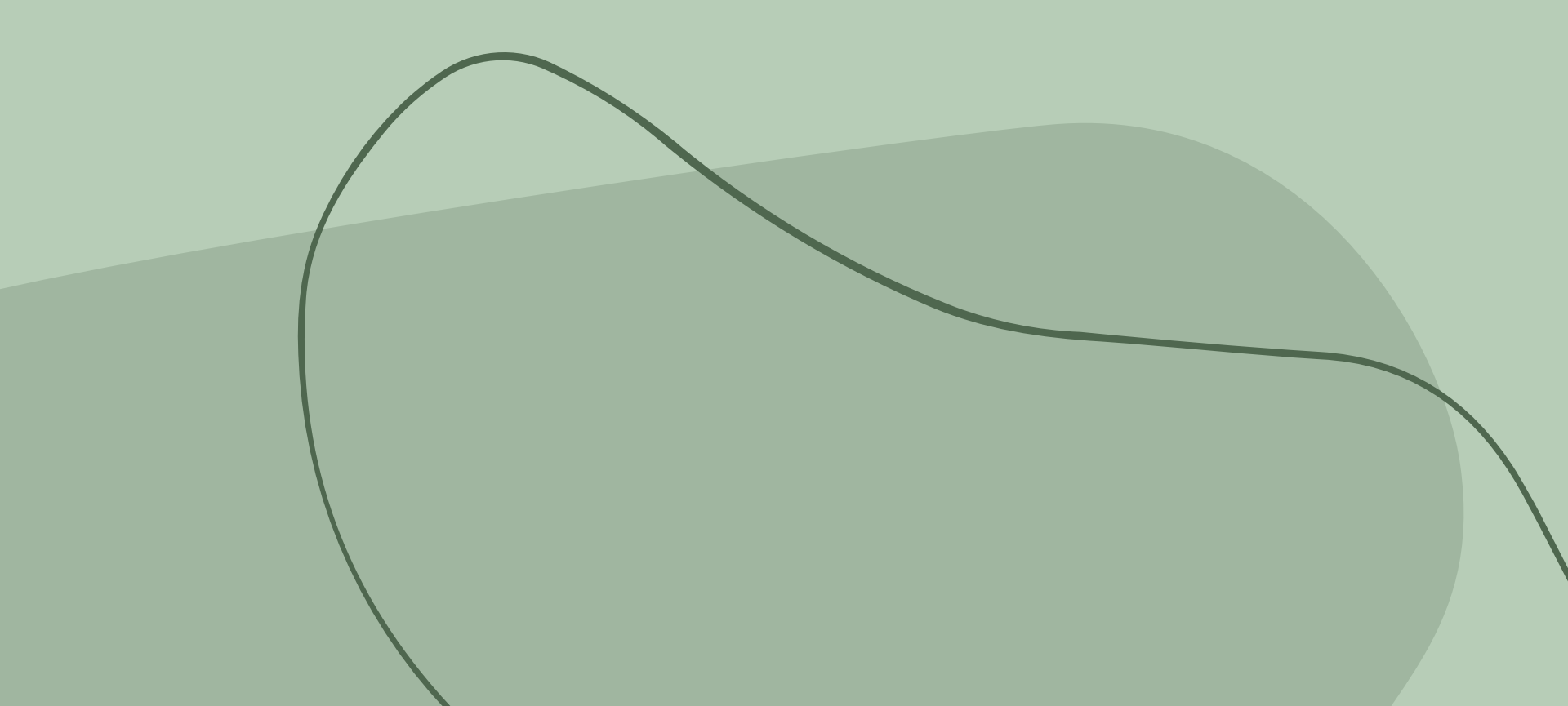


Michaela Scott

- Visits for updates
- Sees real-time information on projects
- Impressed by the leaderboard
- Interested in staying engaged
- Shares the website with her friends

Requirements

Functional Requirements



User Account Management

allowing users to register, log in, log out and manage their account

Educational Resources

visual and interactive elements displaying information about the Foundation's mission and initiatives

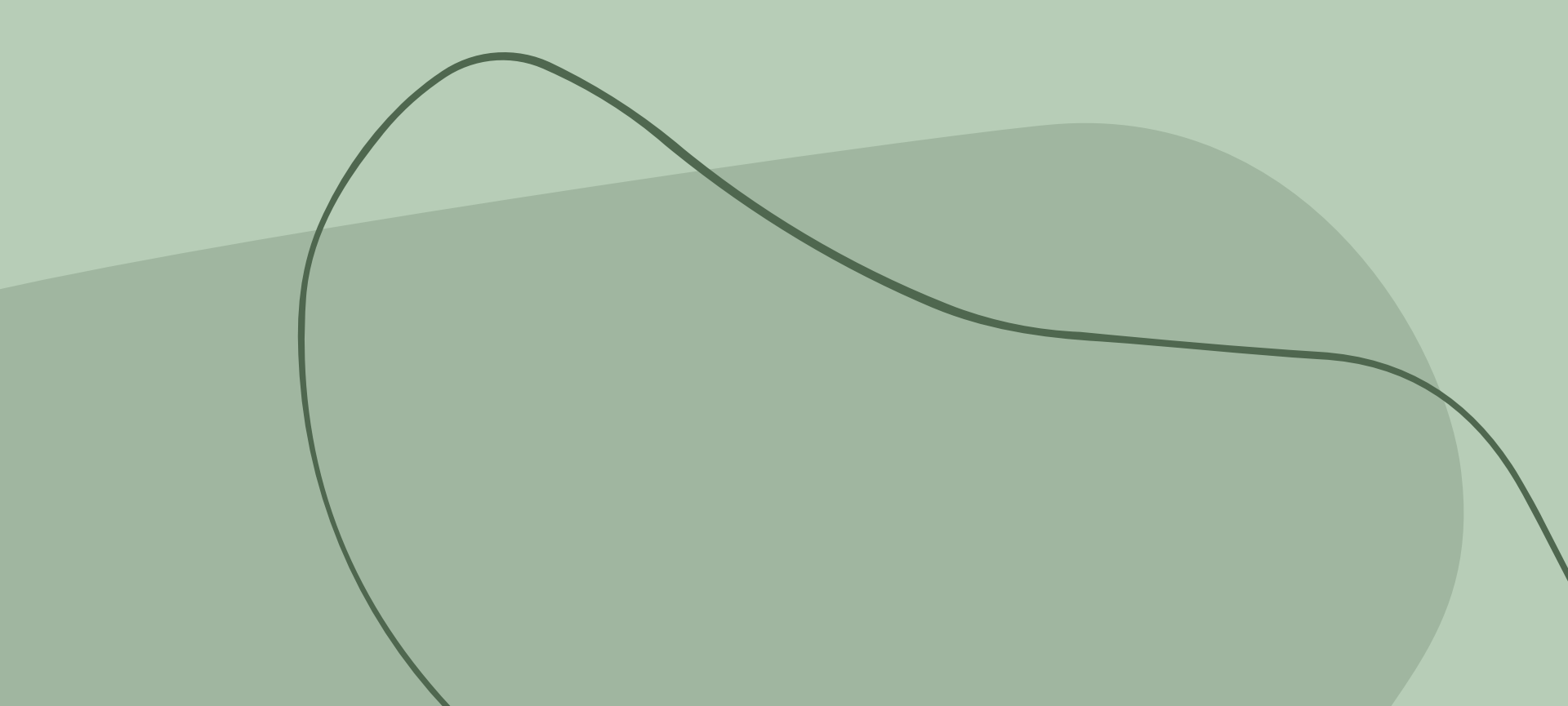
Leaderboard

live and filtered feature tracking user donations to create a compelling call to action

Donation Management

donation forms and payment options that facilitate user financial contributions

Functional Requirements



Project/Event Management

project cards contained on a dedicated webpage with summarised information

Community Engagement

social media links to encourage user engagement across multiple platforms

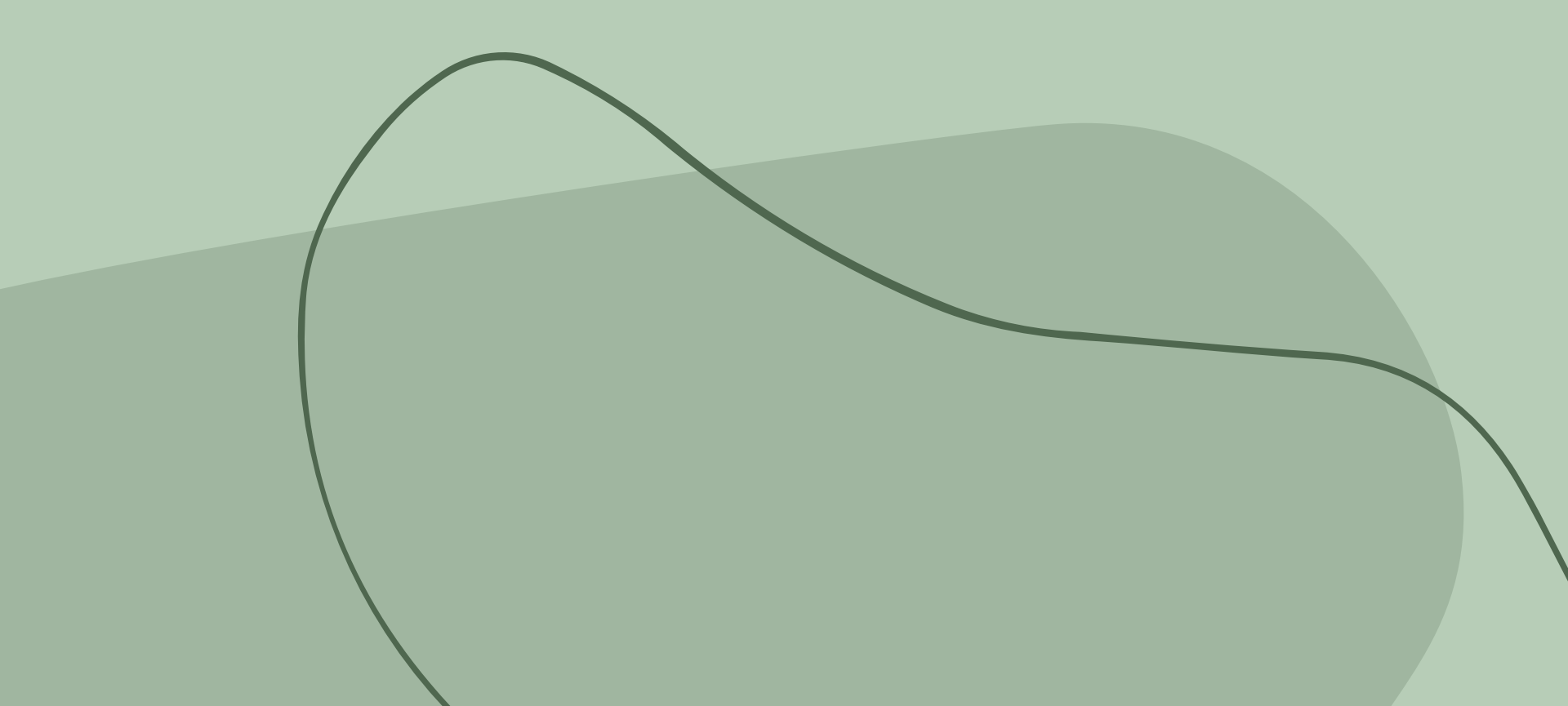
Accessibility Features

text subtitles, video captions, transcripts for audio content, visual notifications and sign language video options

External Contact

Contact Us form and a Live Chat widget to allow users to contact the Foundation directly to receive help or raise queries

Non-Functional Requirements



Performance

quick loading times and upholding scalability to handle increased traffic

Localisation

multiple language support and adherence to cultural sensitivities

Security

2 factor authentication, strong password policies, data encryption and firewalls

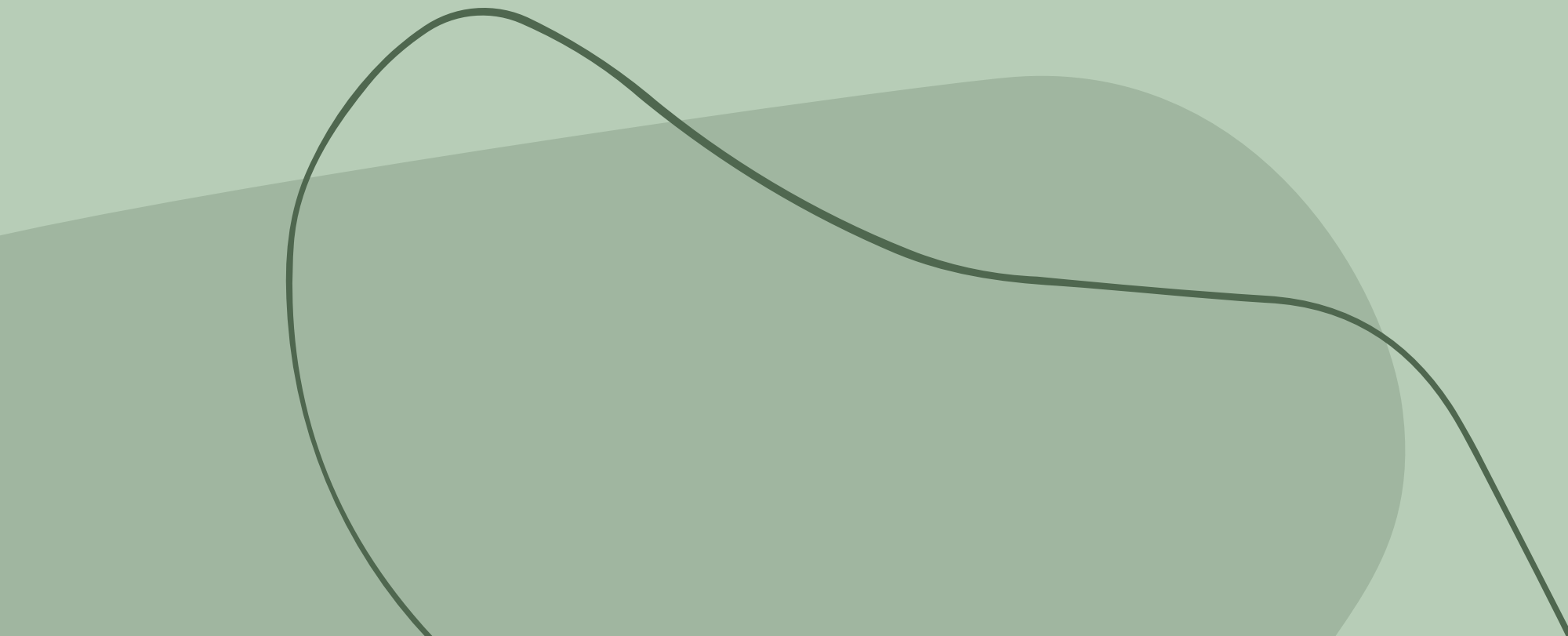
Minimal, Intuitive and Aesthetic Interface

comprehensive site elements reducing textual information, as well as an intuitive and simplistic layout and design

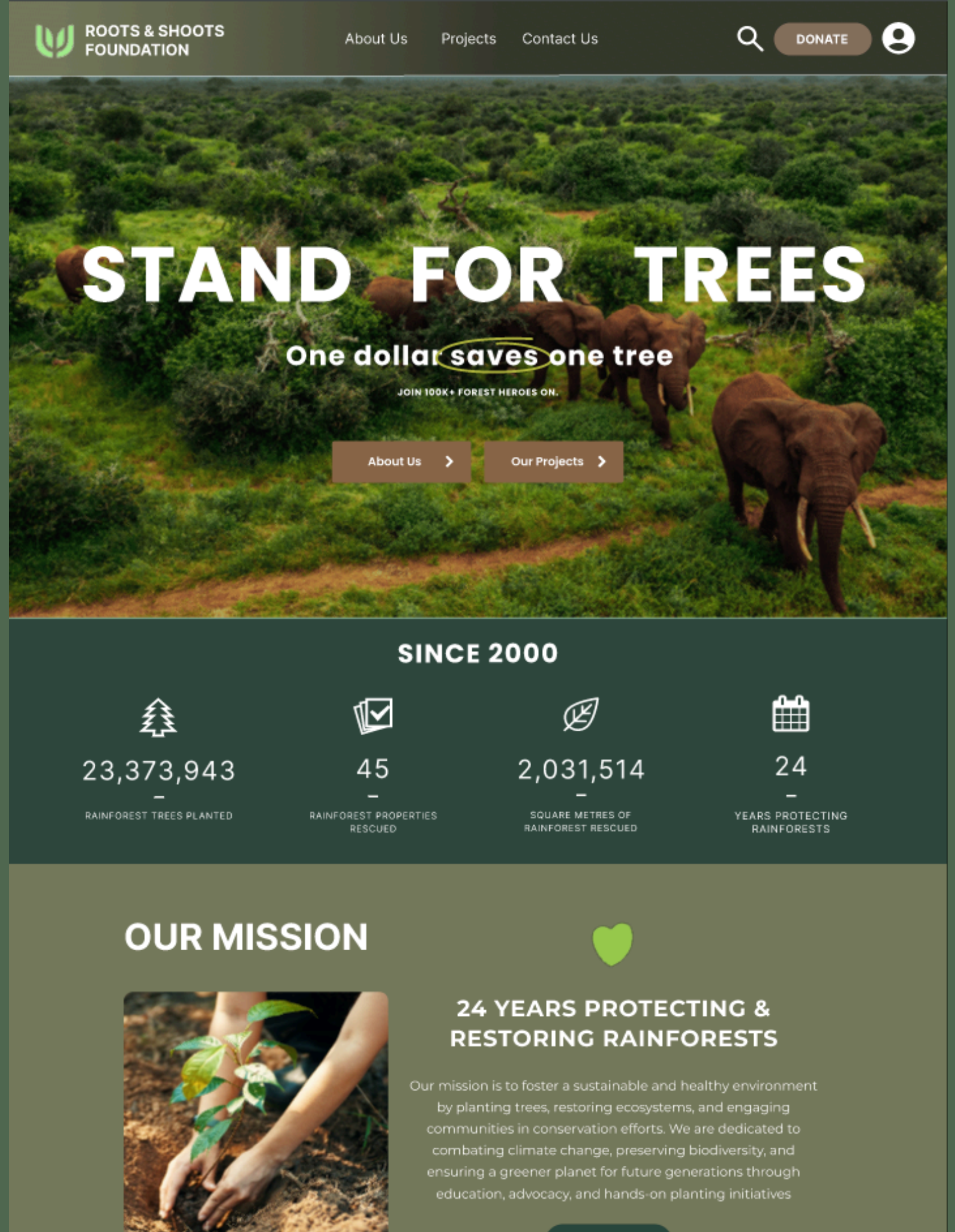
Non- Functional Requirements

Accessibility/Usability

WCAG 2.1 AA supported site features to ensure equal accessibility



Original Prototype





Our mission is to foster a sustainable and healthy environment by planting trees, restoring ecosystems, and engaging communities in conservation efforts. We are dedicated to combating climate change, preserving biodiversity, and ensuring a greener planet for future generations through education, advocacy, and hands-on planting initiatives

> LEARN MORE

OUR PROJECTS



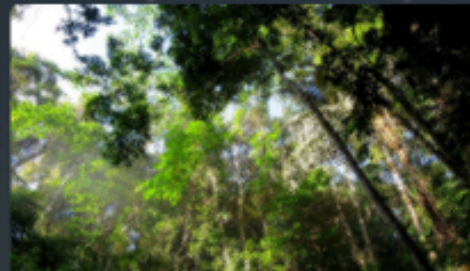
Amazon Rainforest

Illegal logging, agriculture, cattle ranching and mining has resulted in a 17-20% deforestation of the Amazon Rainforest since the 1970s.

🌳 150,000

📍 Peru

LEARN MORE



Eastern Australia

Bushfires, logging and land clearance for agriculture has resulted in an average of 100,000 - 300,000 hectares cleared annually in recent years.

🌳 100,000

📍 Queensland

LEARN MORE



Central Africa

From 2001 - 2020, Cameroon has seen a 11% decrease in forest cover, with 1.6 million hectares lost.

🌳 200,000

📍 Camer

LEARN MORE

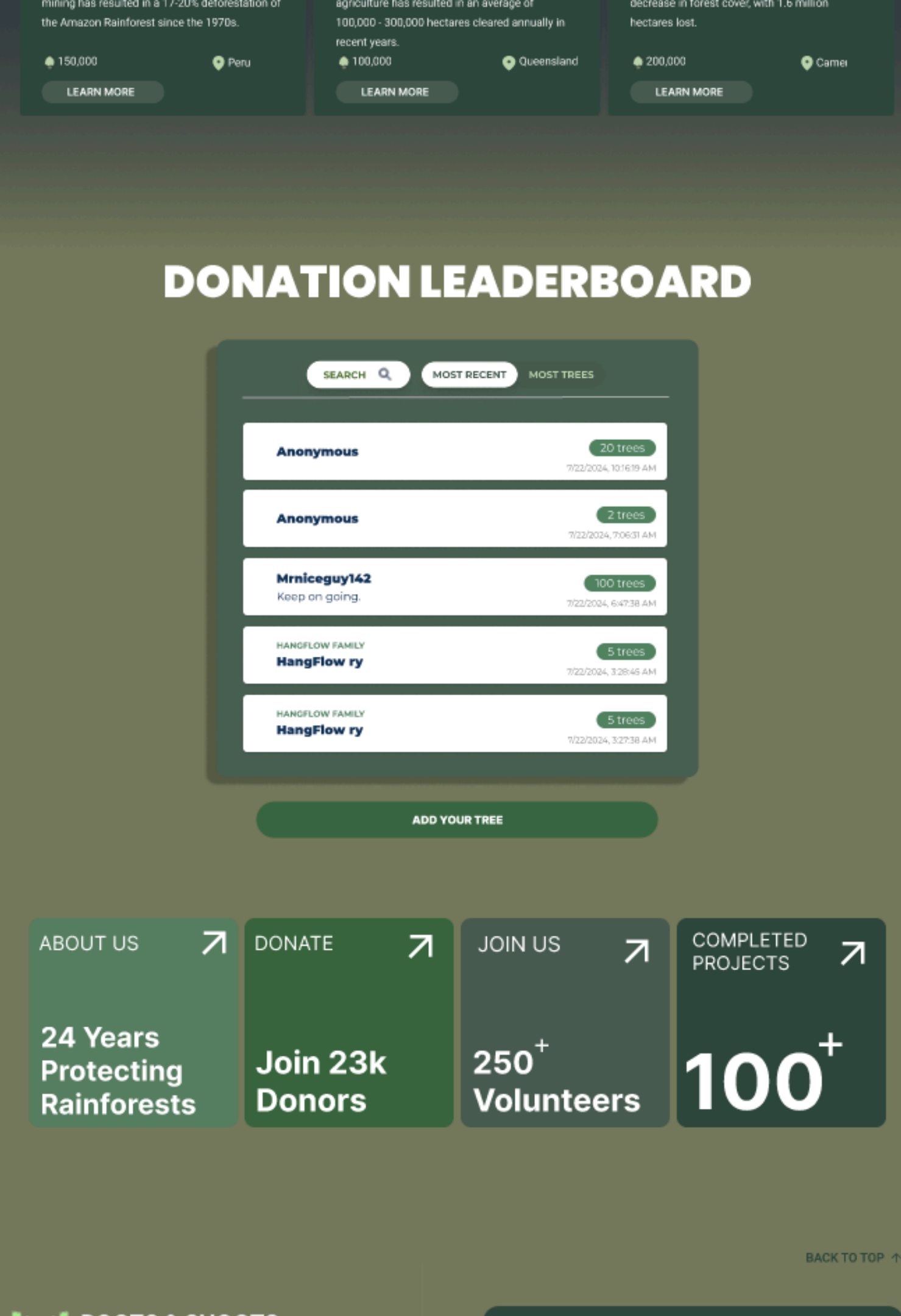
Interactive World Map

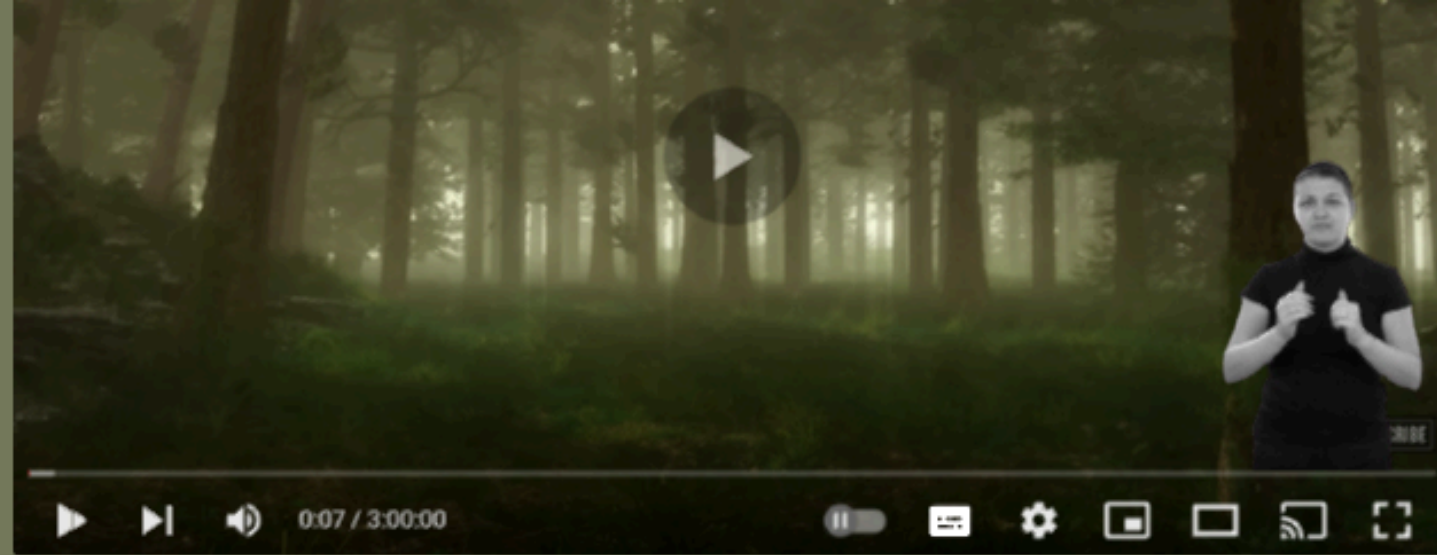
Users are able to drag through a World Map displaying the Foundation's current initiatives in each continent of the world, condensing the information into a visual format that fosters exploration and engagement.

Live Donation Leaderboard

Displaying user contributions in a live Donation Leaderboard to foster a sense of collaboration, competition and create a call to action.

Users are also able to see the Foundation's tangible impacts and stay up-to-date with initiatives





show transcript

sign language interpreter

OUR GOAL:

150,000

TREES

110,034



CO₂

increasing forest cover will promote the Amazon's role as a carbon sink and offset CO2 emissions



replanting will replenish degraded areas of the Rainforest, which can restore biodiversity and stabilize ecosystems



more trees will positively influence local precipitation patterns and maintain humidity levels globally and regionally



planting native tree species can provide a habitat for a wide range of animal and plant species

BACK TO TOP ↑

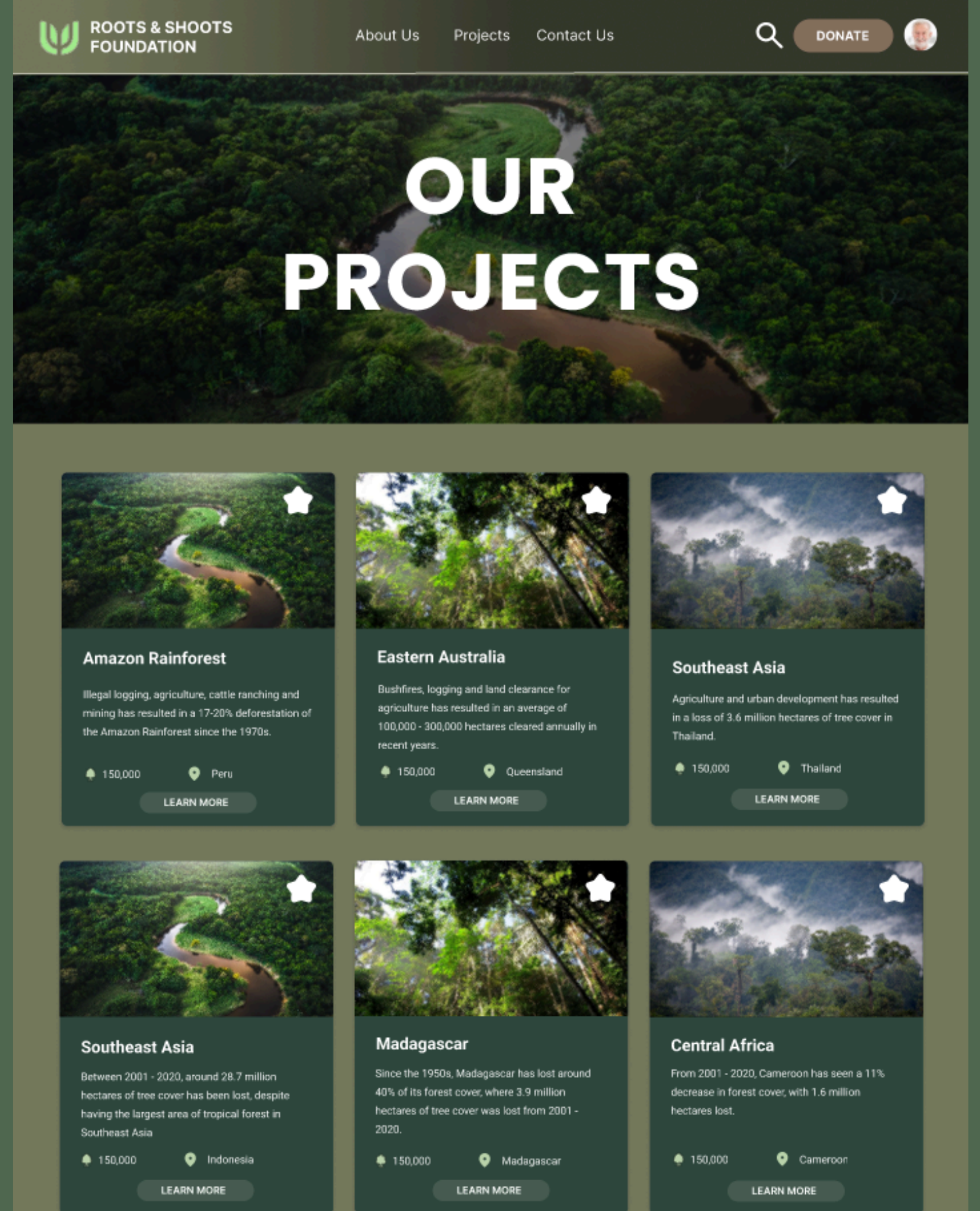
Live Progress Tracker

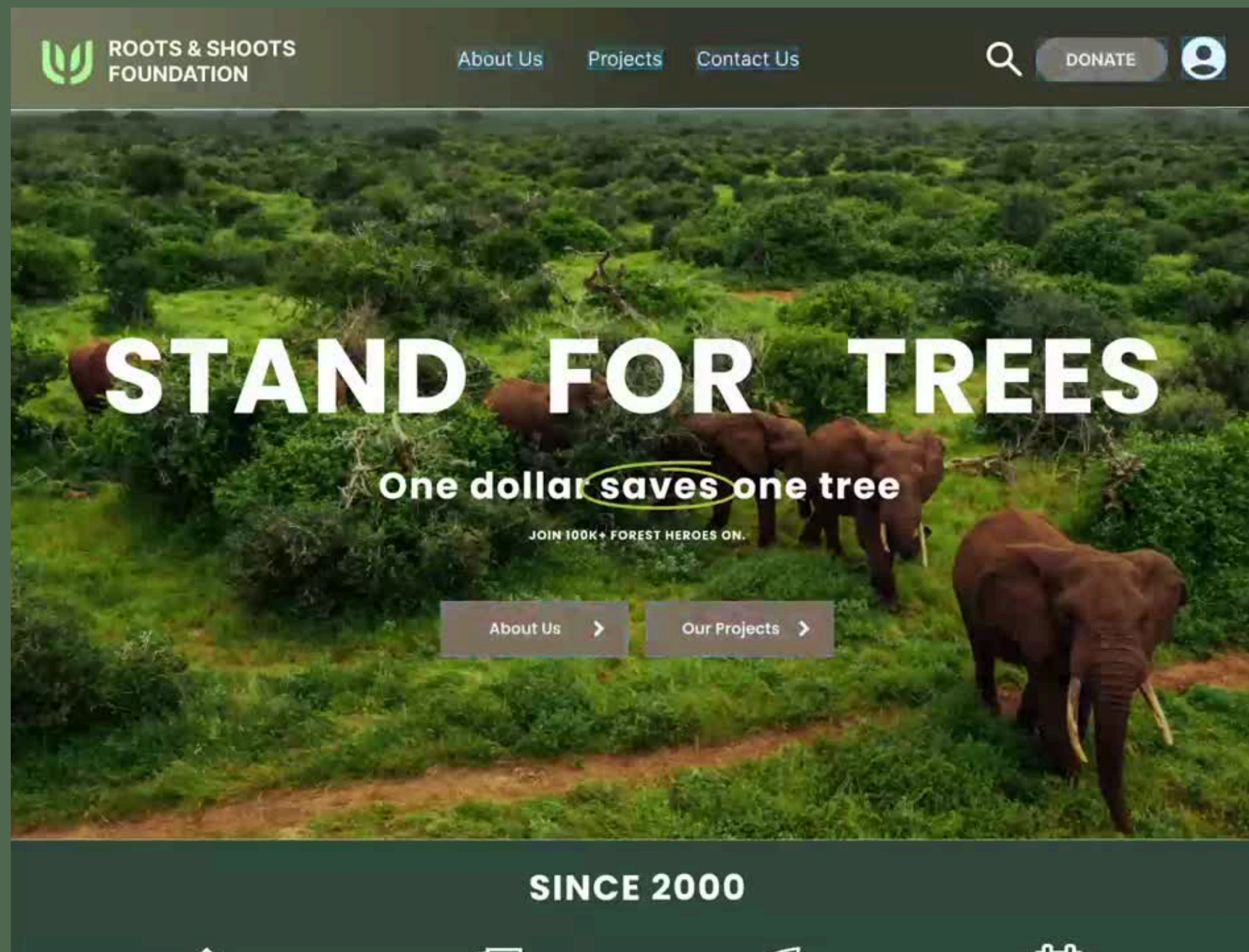
Users are able to view a live Progress Bar that displays the current trees planted for the Project. This visual presentation allows users to glean the progress towards the final goal in an engaging and dynamic manner

Projects Page

The Projects are ordered succinctly in Project Cards resembling game missions and objectives. This contributes to a simplistic and minimalistic interface that will engage users.

Project cards can also be favourited allowing users to easily access Projects and receive updates about the progress of the Project

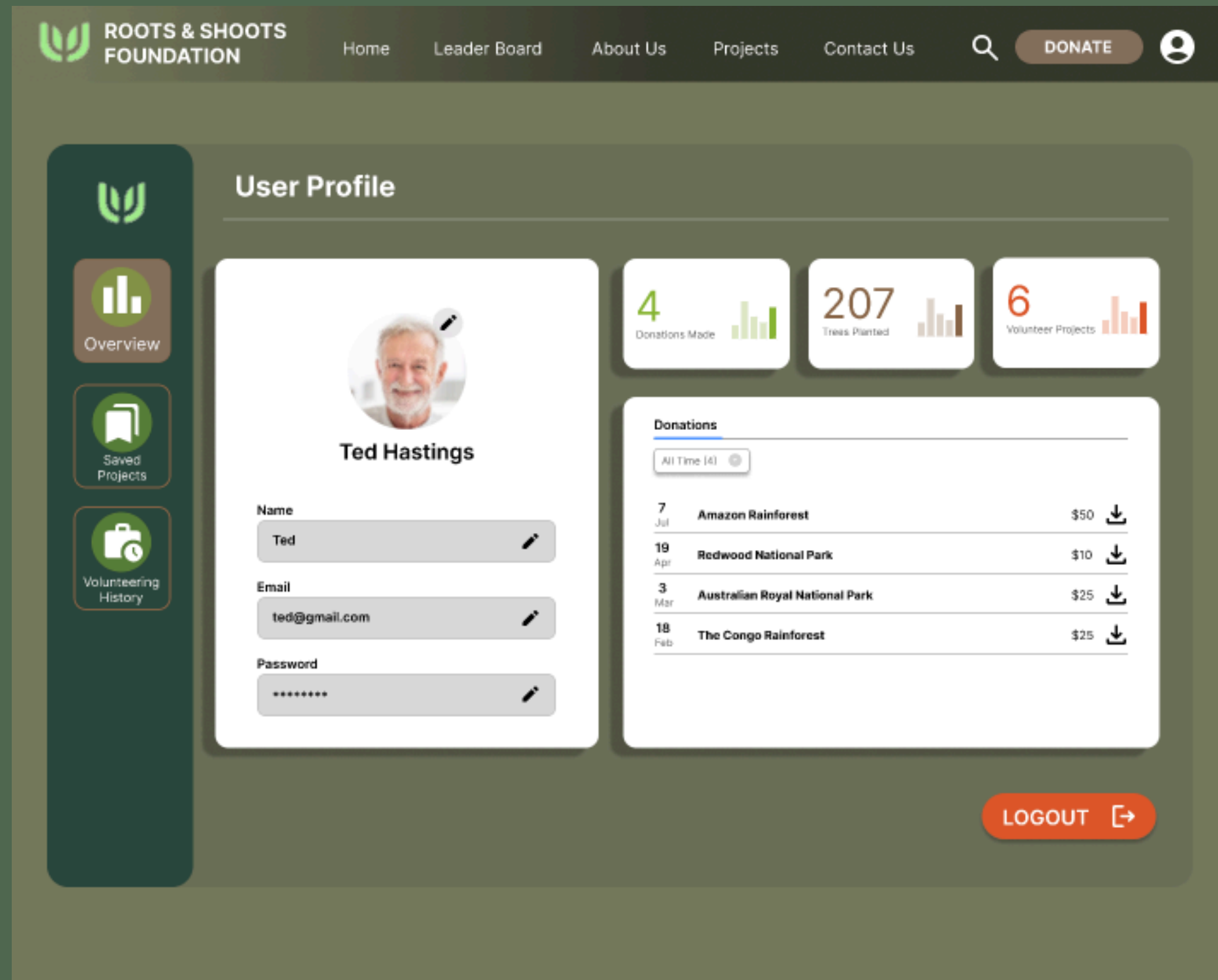




Donation Forms and Payment Options

Donation Forms can be accessed via the Donate button, with a prefilled and minimalistic Donations Form to minimise user action. Multiple payment options are also provided.

The Donations confirmation page will also explicitly state the usage of the donation towards planting a specific number of trees in a specific region of the world



User Account Management

Users are able to register, log in, log out and change profile settings through a simple interface.

Key information such as Donation History, Saved Projects and Volunteering History allow users to quickly access resources related to their account activity, thereby increasing user efficiency



Usability Evaluations

Userbility Testing Results

Reference	Issue	Priority	P1	P2	P3	P4	P5	P6	P7	P8	NOP
1	The absence of a dedicated Home page redirection feature made it difficult for users to navigate back to the landing page, resulting in user confusion	4	x				x				2
2	Lack of exit breadcrumb made it confusing for users to glean their location on the site and retrace their steps	3									N/A
3	Donation Confirmation page lacked clear instruction or button to navigate to the next page, making it difficult for users to progress with Payment	2	x	x		x					3
4	Projects page lacked ability to filter Projects, resulting in inefficient user workflow as they had to scroll through every Project to search for their desired target	3						x	x		2
5	Donations Confirmation Page lacked adequate overview of information to allow users to double check their entry before proceeding this purpose.	4	x	x	x		x				4
6	Contact Form lacked sample responses in entry fields to guide users on the correct format of the entries, which can potentially lead to wrong inputs	2									N/A
8	The lack of a dedicated button for the leaderboard made it hard for users to find and navigate it.	2	x	x		x		x	x		3
9	There is only one Donations button throughout the website located on the navigation bar, which may be easily missed by users.	2						x	x		2
10	The world map had no indication it was draggable, and users were not sure if they could scroll through.	2		x			x	x	x		4
11	Project cards on the “Saved Profile” page are different to cards in the “Projects” page causing inconsistencies.	2						x			1
12	The navigation bar is too wide, where it is infringing on the visibility of the rest of the page, making it difficult for users to process information quickly	1						x			1
13	The blurb on the donations page is too long and hides the thank you message at the bottom of the paragraph.	1						x	x		2
14	Saved Projects and volunteering sections on the user profile page have a visible but non-functional scroll bar.	1						x			1
15	The contrast ratio of the text in the leaderboard toggle wasn’t sufficient which minimized its visibility.	3									N/A
16	The Projects page lacked a filter, which can result in increased user workflow if a user is seeking a specific Project	2									N/A

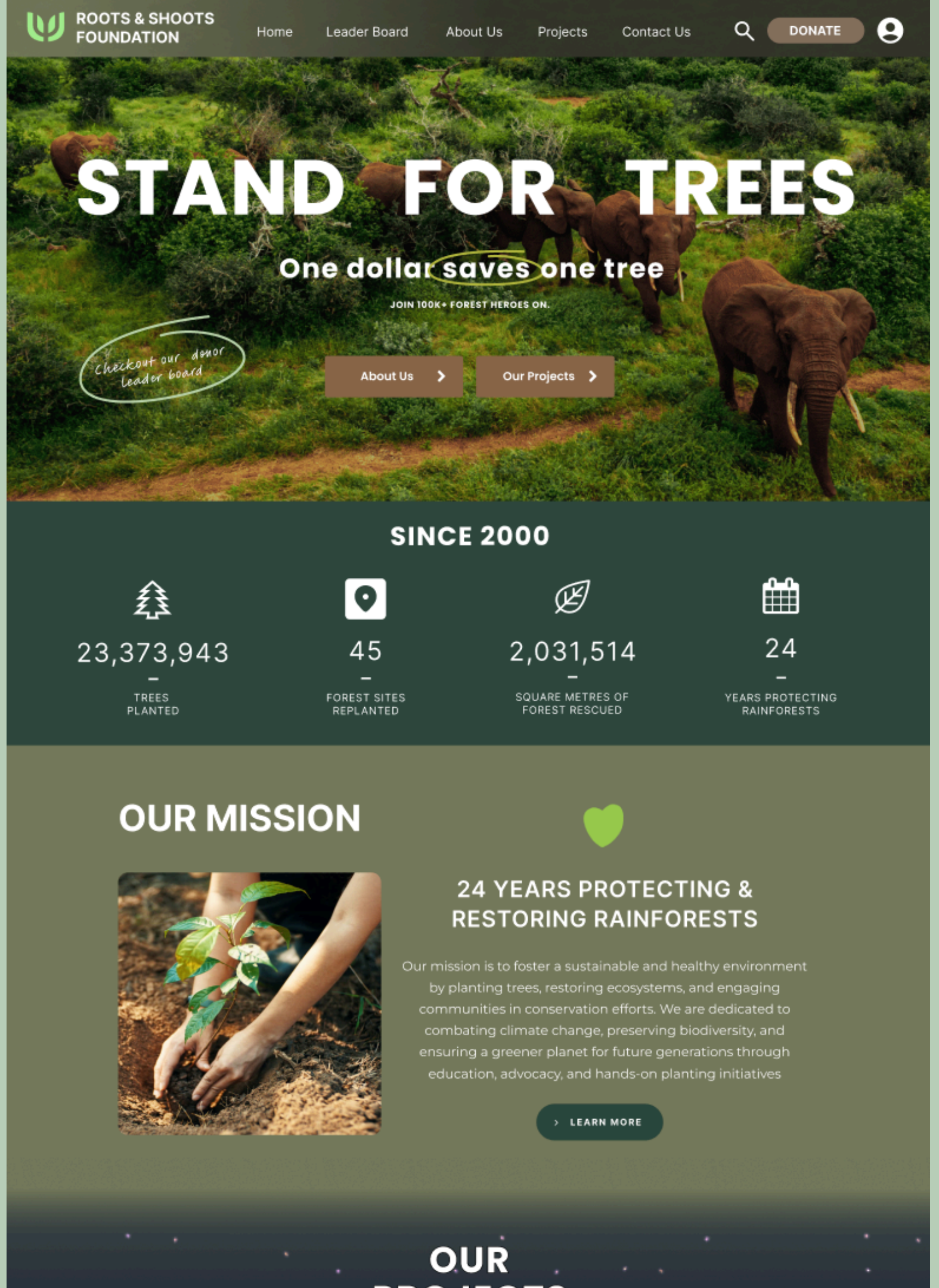
1 = mini issue, 2 = minor issue, 3 = major issue, 4 = showstoppe

Highest Priority Issues

Reference	Issue	Priority	P1	P2	P3	P4	P5	P6	P7	P8	NOP
1	The absence of a dedicated Home page redirection feature made it difficult for users to navigate back to the landing page, resulting in user confusion	4	x				x				2
2	Lack of exit breadcrumb made it confusing for users to glean their location on the site and retrace their steps	3									N/A
4	Projects page lacked ability to filter Projects, resulting in inefficient user workflow as they had to scroll through every Project to search for their desired target	3						x	x		2
5	Donations Confirmation Page lacked adequate overview of information to allow users to double check their entry before proceeding this purpose.	4	x	x	x		x				4
8	The lack of a dedicated button for the leaderboard made it hard for users to find and navigate it.	2	x	x		x		x	x		3
10	The world map had no indication it was draggable, and users were not sure if they could scroll through.	2		x			x	x	x		4
11	Project cards on the “Saved Profile” page are different to cards in the “Projects” page causing inconsistencies.	2						x			1
15	The contrast ratio of the text in the leaderboard toggle wasn’t sufficient which minimized its visibility.	3									N/A

1 = mini issue, 2 = minor issue, 3 = major issue, 4 = showstoppe

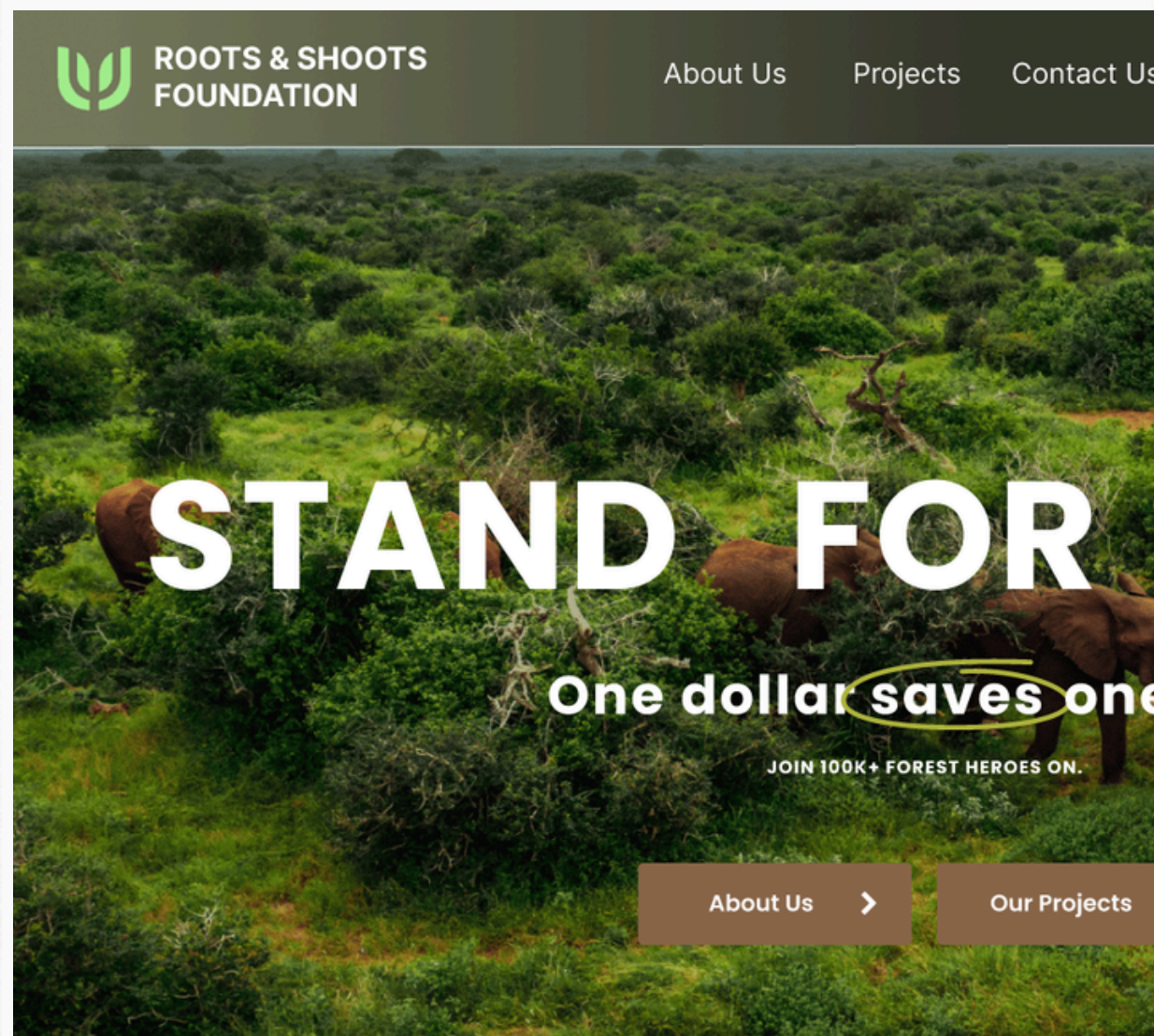
Revised Prototype



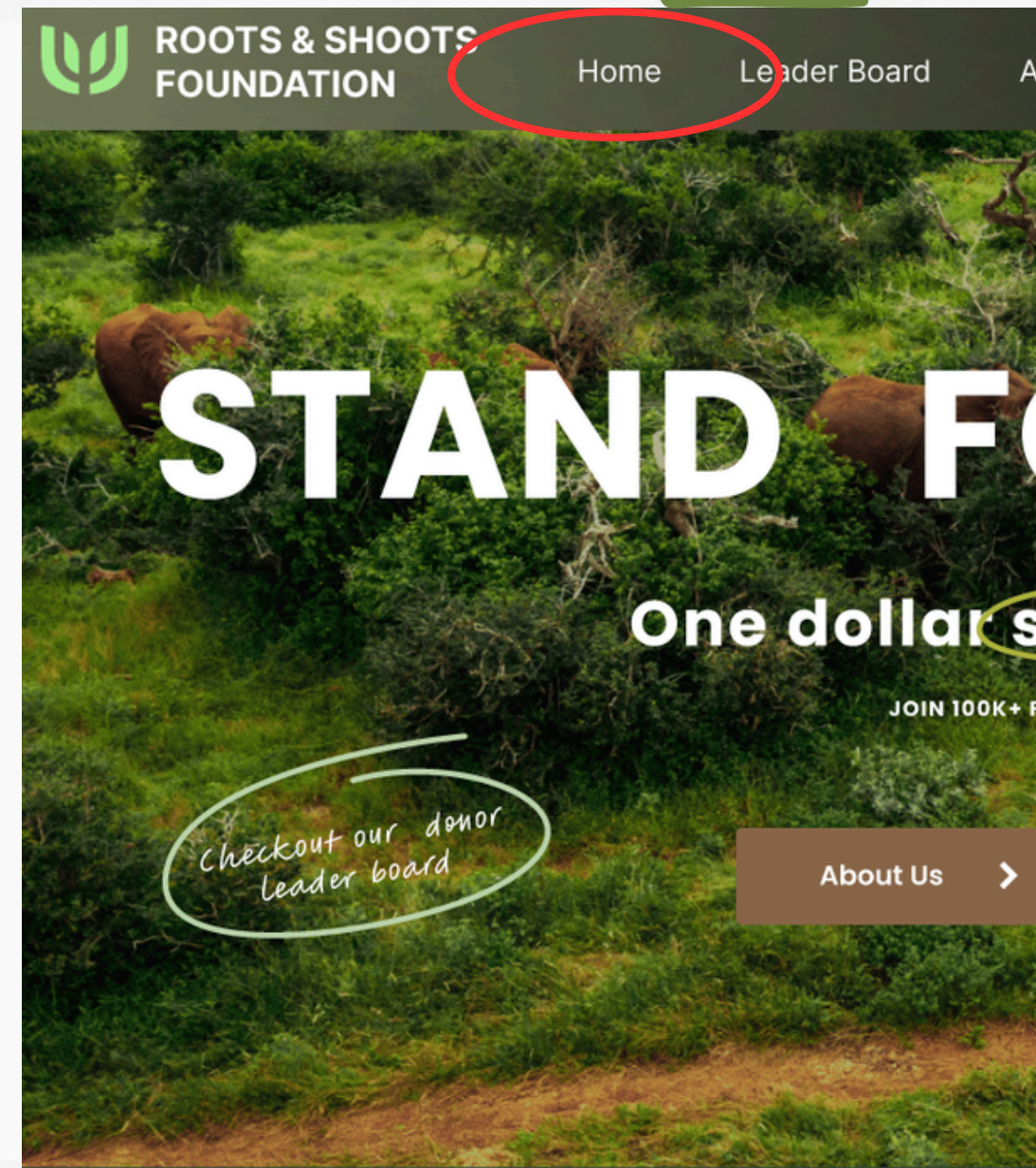
Issue Ref #1

The absence of a dedicated Home page redirection feature made it difficult for users to navigate back to the landing page, resulting in user confusion

- We went in with the assumption that every user would click on the pages' logo to return home.
 - However, this only catered towards highly experienced web users
- Affects **Efficiency** and **Learnability**; Users take longer to figure out how to navigate home and learn that the logo can navigate them home.
- Breaches principles of **Consistency and Standards** as most pages have a Home navigation button, as well as **visibility** as main features are not visible to users.
- Creates a **frustrating** and **confusing** user experience.



before

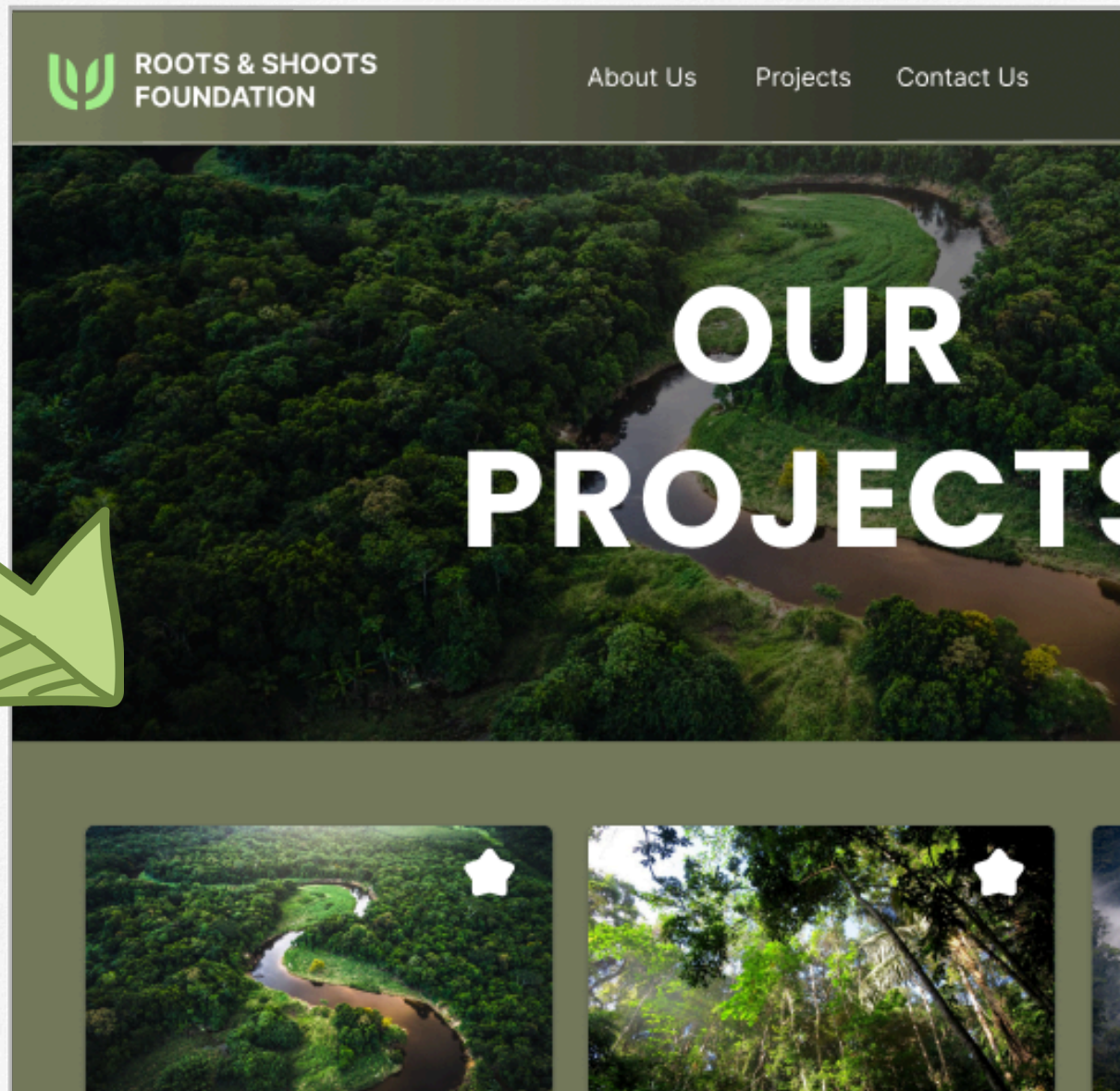


after

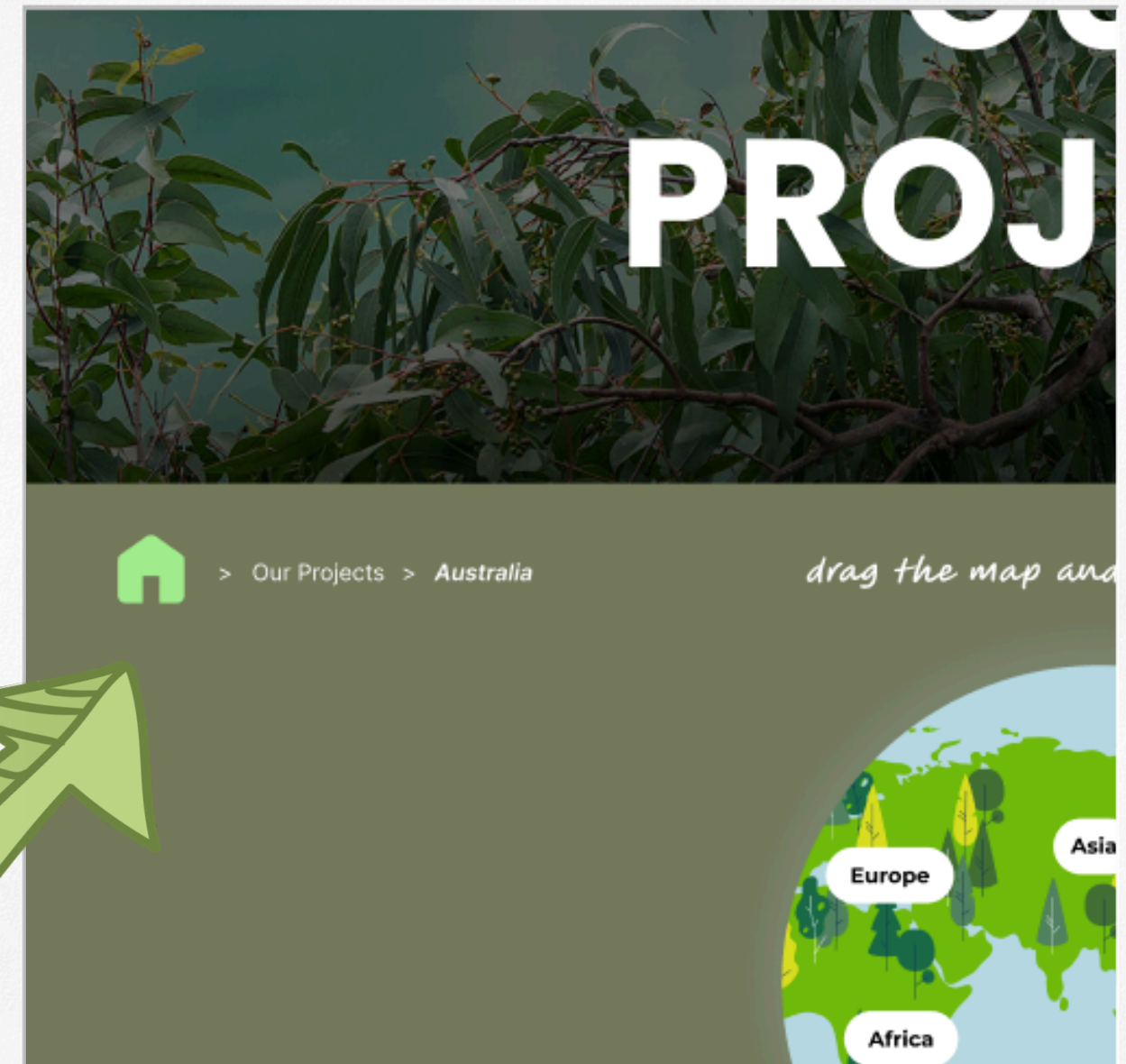
Issue Ref #2

Lack of exit breadcrumb made it confusing for users to glean their location on the site and retrace their steps

- Users can get nested towards a certain page with no way to navigate backwards through the projects.
- Breaches **User Control and Freedom** as users have no way to navigate back to their previous page with features presented to them.
- Affects **efficiency** as users require more steps to trace backwards, as well as **learnability** as it takes users longer to understand the logic to navigate around the site.
- Creates a **confusing and frustrating** experience.



before

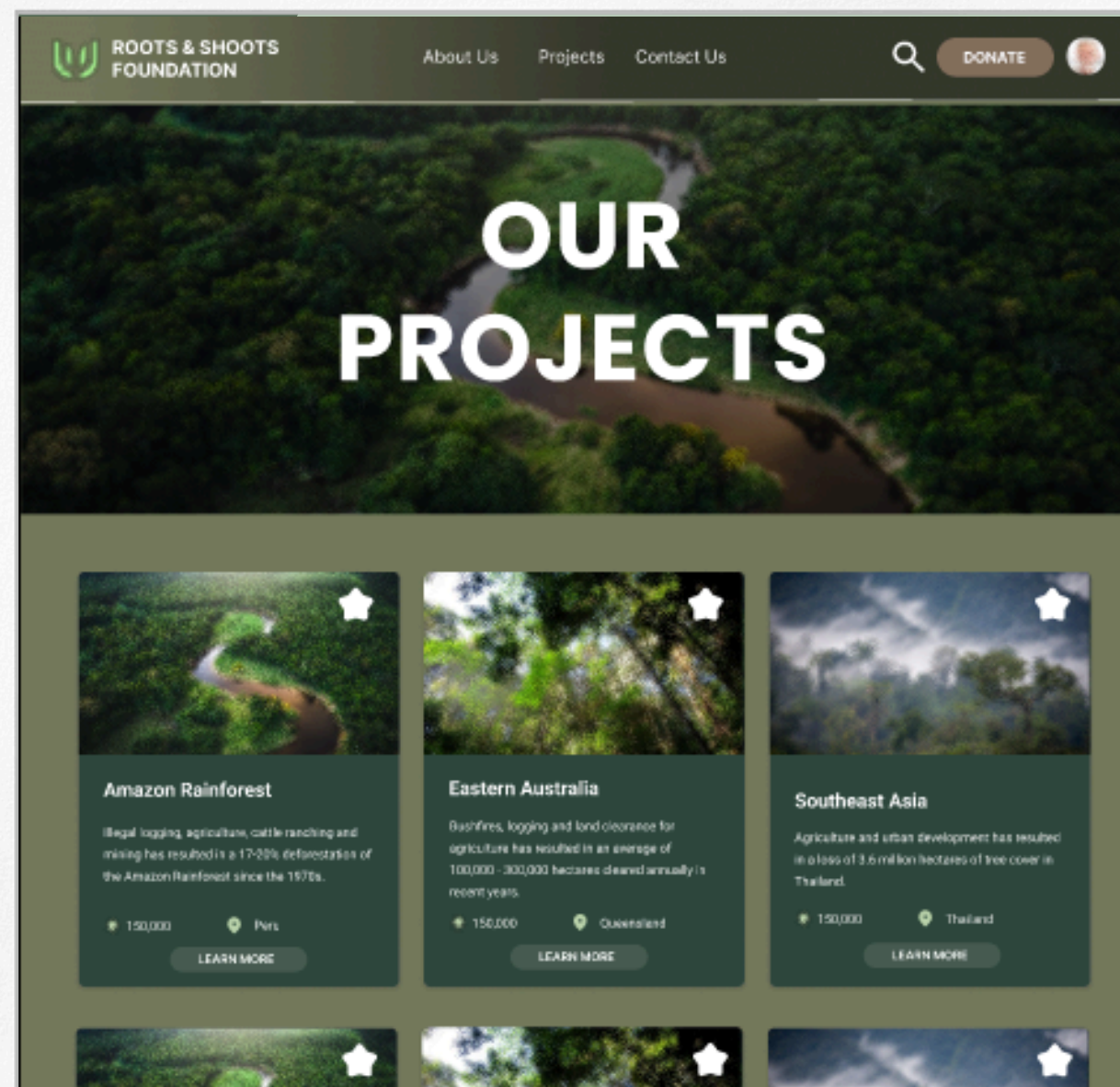


after

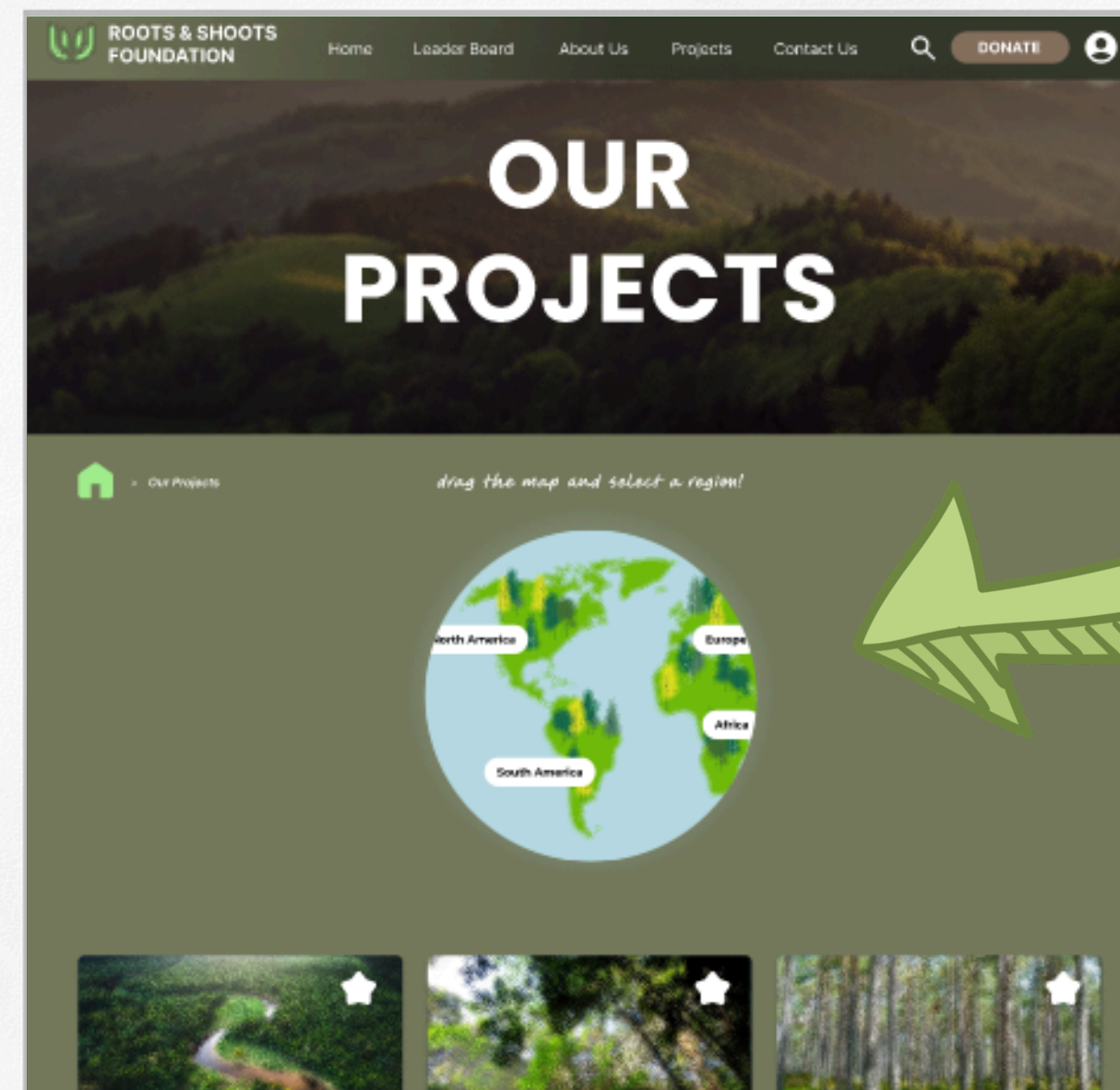
Issue Ref #4

Projects page lacked ability to filter Projects, resulting in inefficient user workflow as they had to scroll through every Project to search for their desired target

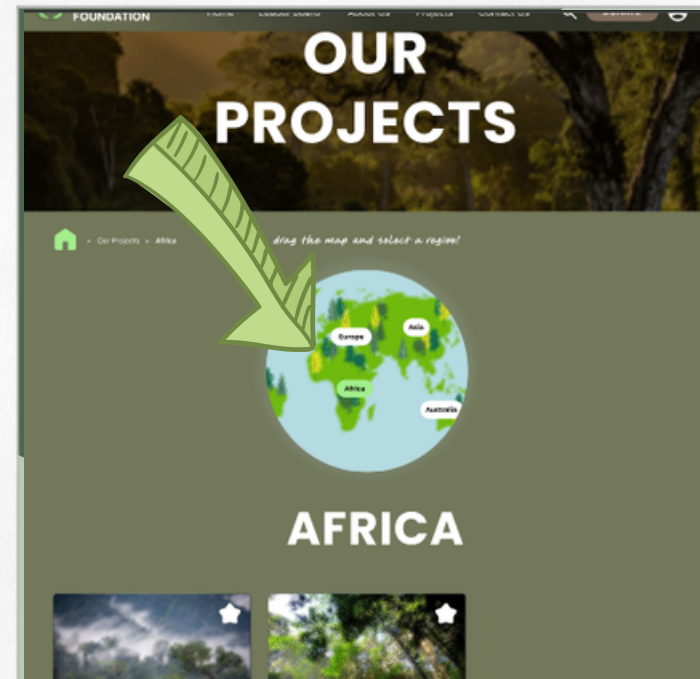
- A lack of a filter affects the scalability of the site as projects become increasingly harder to navigate without a filter.
- Reduces **Flexibility and Efficiency of Use** as tools are not provided for users to effectively filter out projects they want to view
- Affects **efficiency** as users take longer to find the project they want; **utility** as the site lacks the function for users; **effectiveness** as the projects page is not effective in helping users find the project they want
- Creates a slow and **unpleasant** user experience



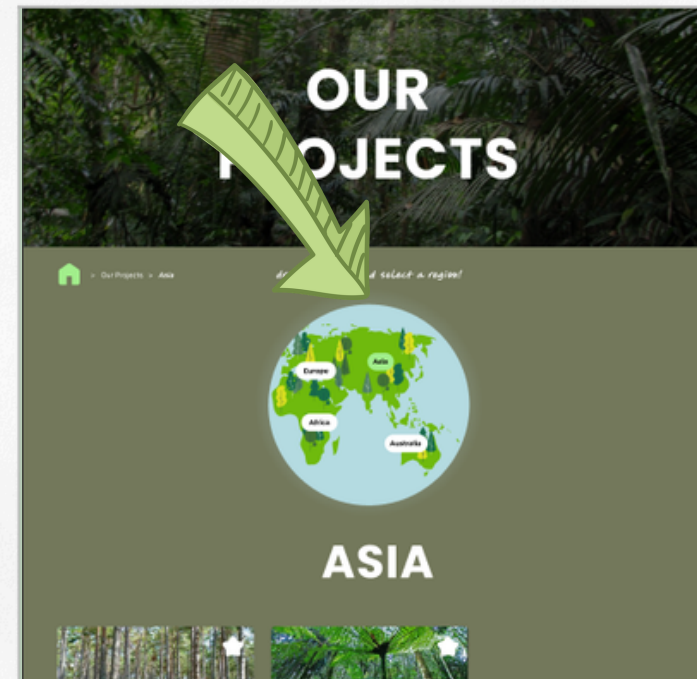
before



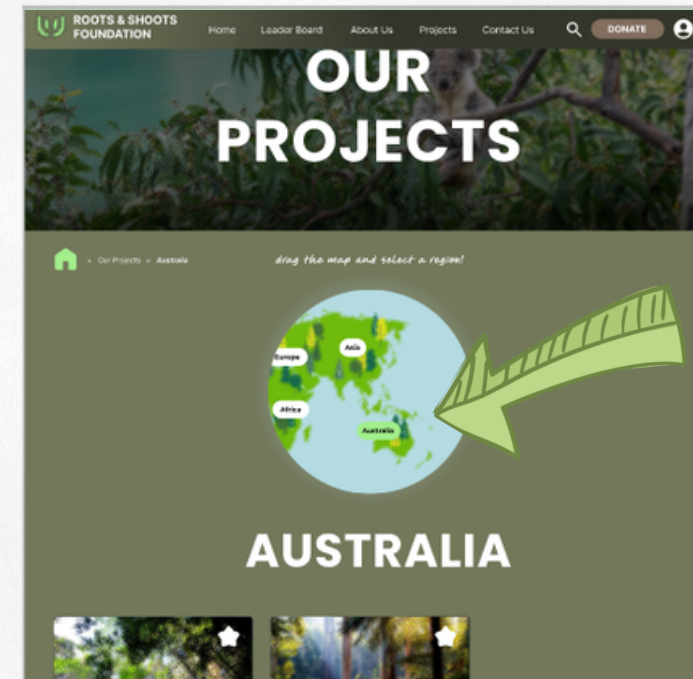
after



after



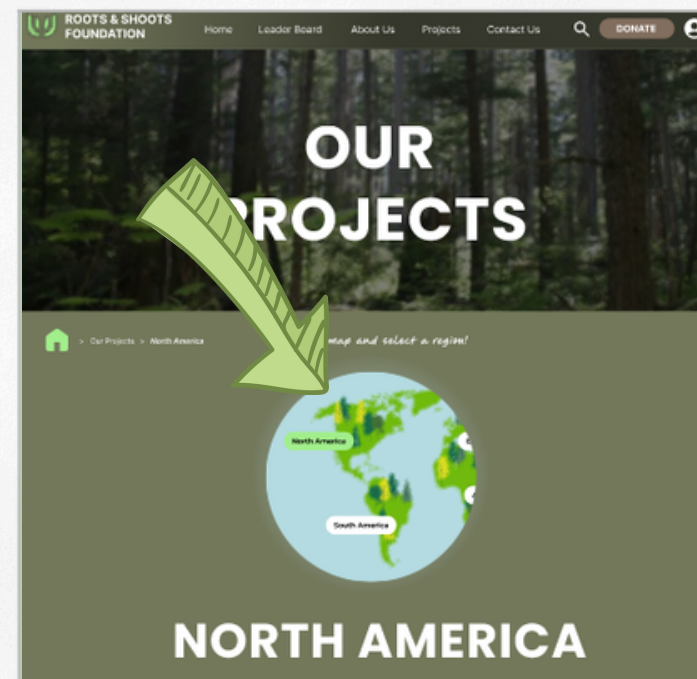
after



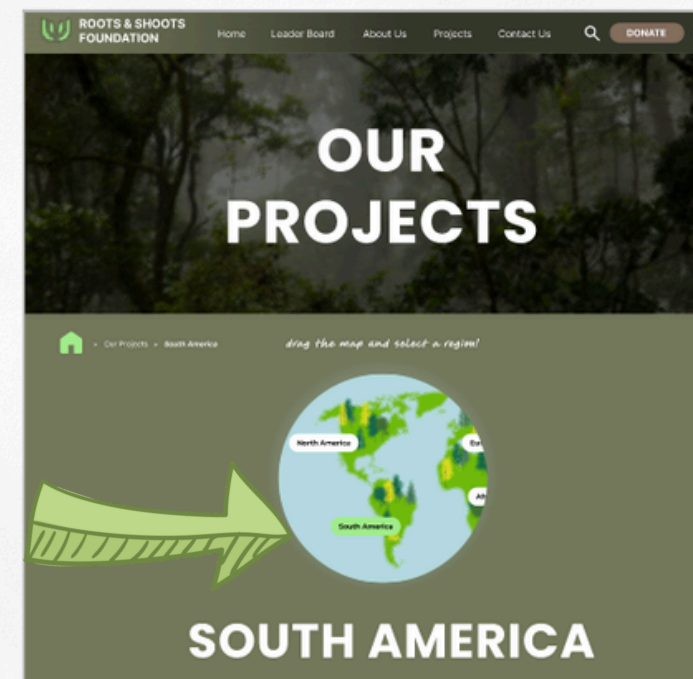
after



after



after



after

Issue Ref #5

Donations Confirmation Page lacked adequate overview of information to allow users to double check their entry before proceeding this purpose.

- Users want to be able to confirm their inputted details before doing a sensitive action such as donating and spending their money
- The lack of **Error Prevention** of their details before pressing the “donate” button is against design principles
- It affects the **efficiency** as users spend more time navigating back and forth their details and confirm page, as well as the **effectiveness** of the confirmation page as users have to ways to actually “confirm” their details
- Creates a **confusing** and **frustrating** user experience.

Total Amount:

\$XX.XX

Confirm PayPal Donation

OPEN IN NEW TAB

← GO BACK

ROOTS & SHOOTS
FOUNDATION

Subscribe for updates

First Name

Last Name

BACK TO

before

Donation Details

Project Choice

Amazon Rainforest

Payment Method

☒ Credit/Debit Card

☐ PayPal

☐ Apple Pay

Donation Name (to be displayed)

Anonymous

Message (to be displayed)

No Message

Total Amount:

\$XX.XX

Confirm Donation Amount

PROCEED →

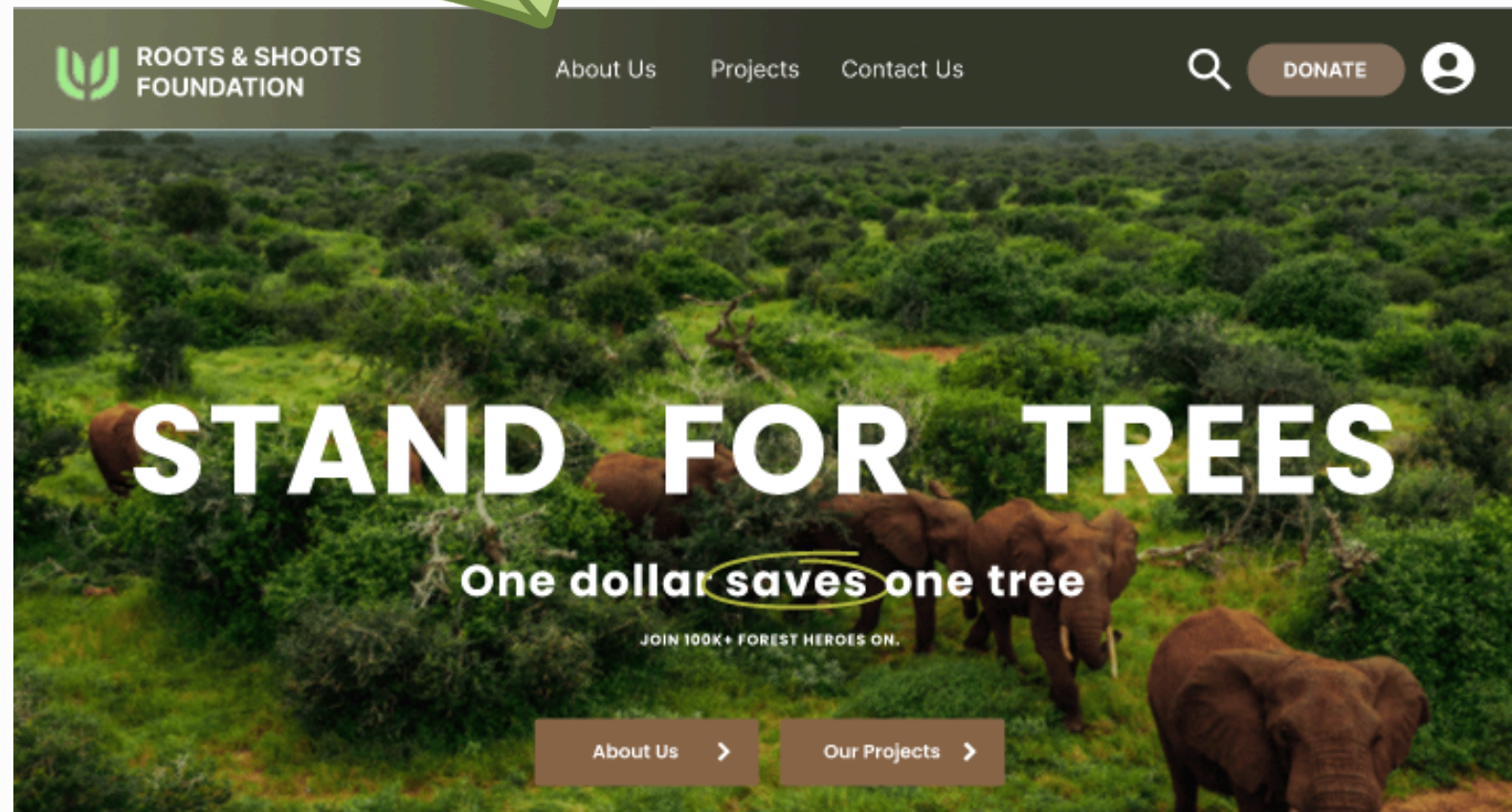
← GO BACK

after

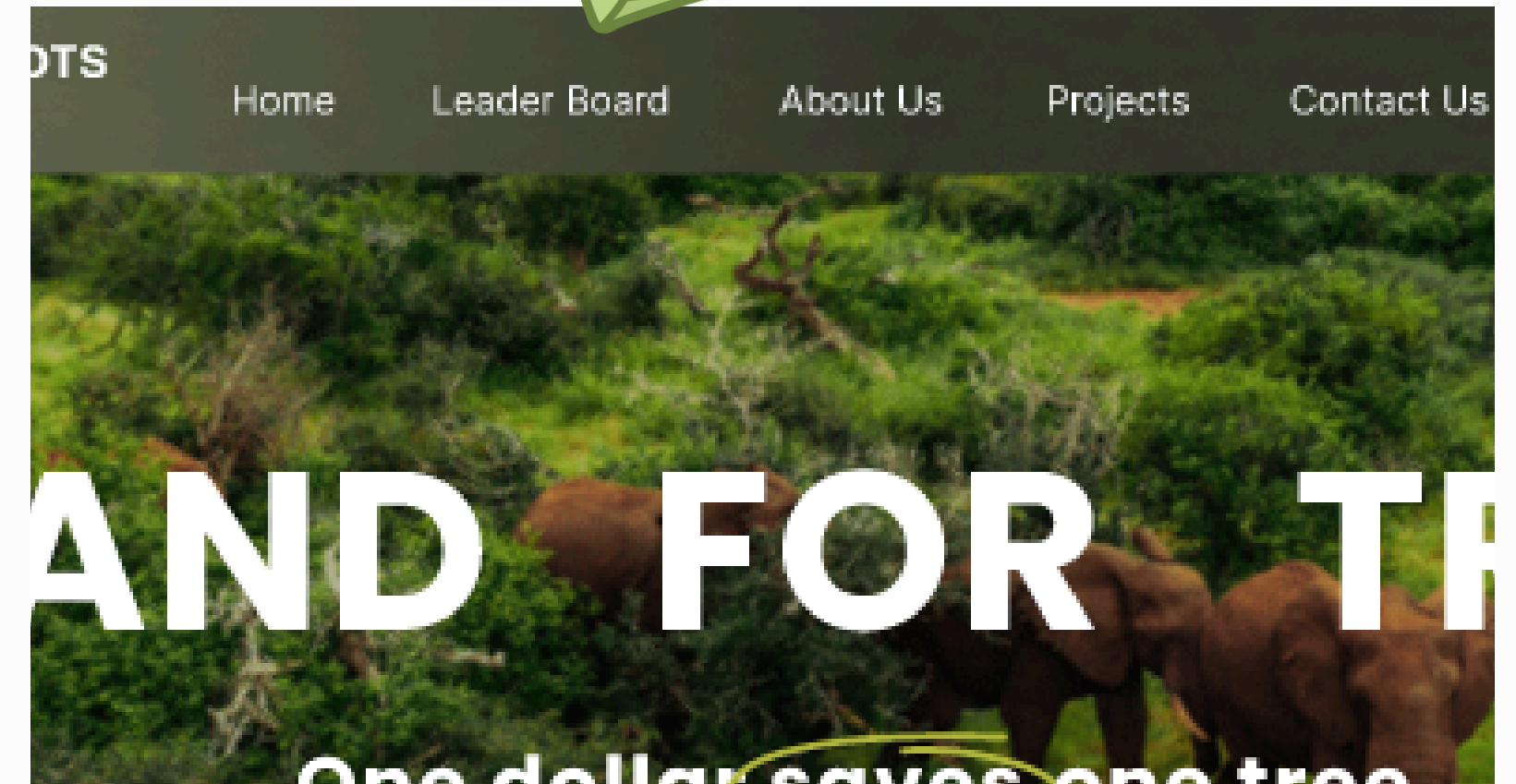
Issue Ref #8

The lack of a dedicated button for the leaderboard made it hard for users to find and navigate it.

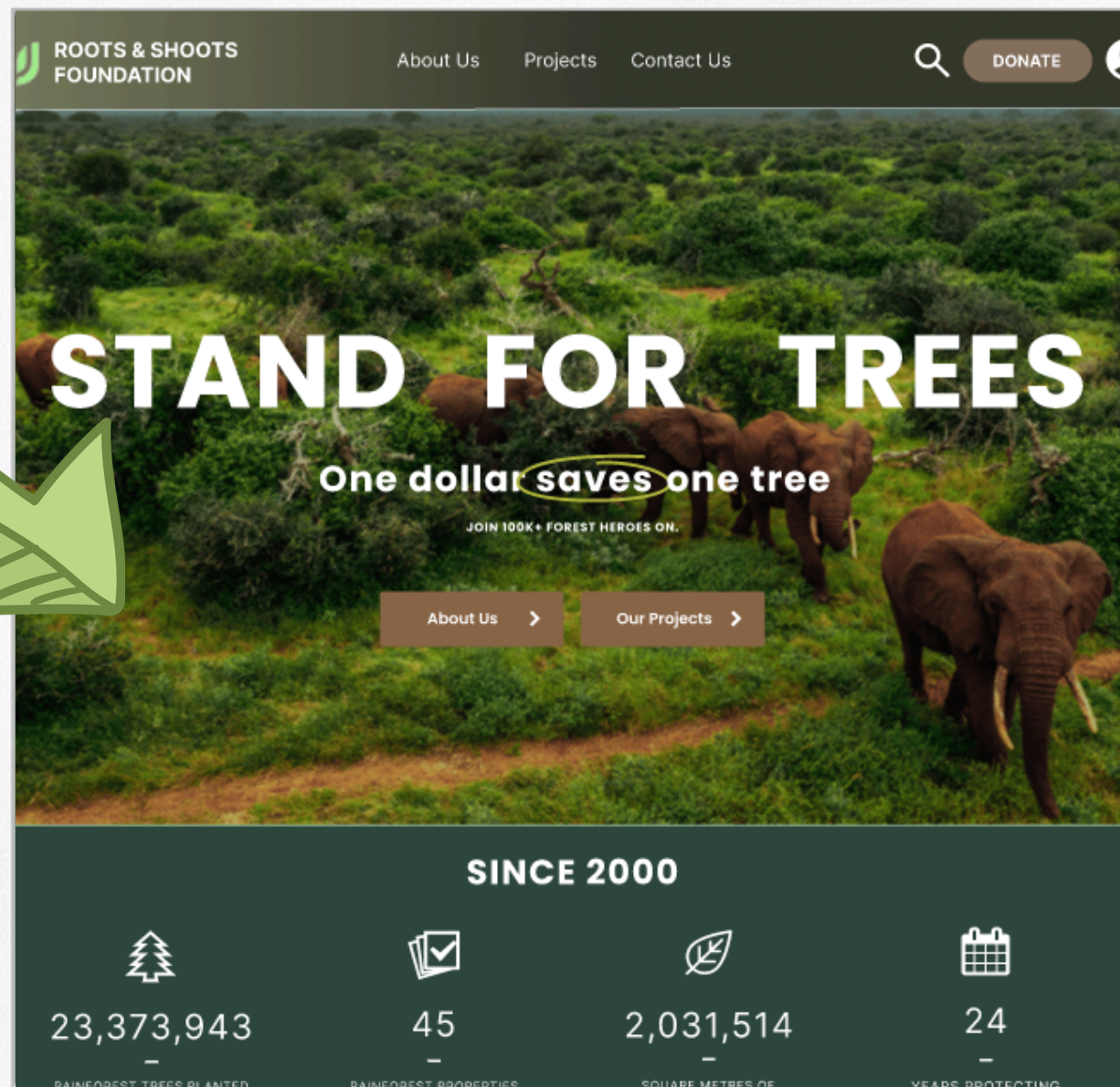
- A lack of a **leaderboard navigation button** is frustrating as one of the main features of the site is to view this leaderboard
- Reduces **visibility** as the sites' main feature of a leaderboard is not clearly shown to the users eyes, and thus users are unable to ascertain the main functionality of the site
- Affects the **learnability** and **efficiency** of user actions as users on average take much longer time to figure out where the leaderboard is, and find it hard to learn this information.
- Creates a **confusing** experience that makes **users feel stupid**.



before



after



before

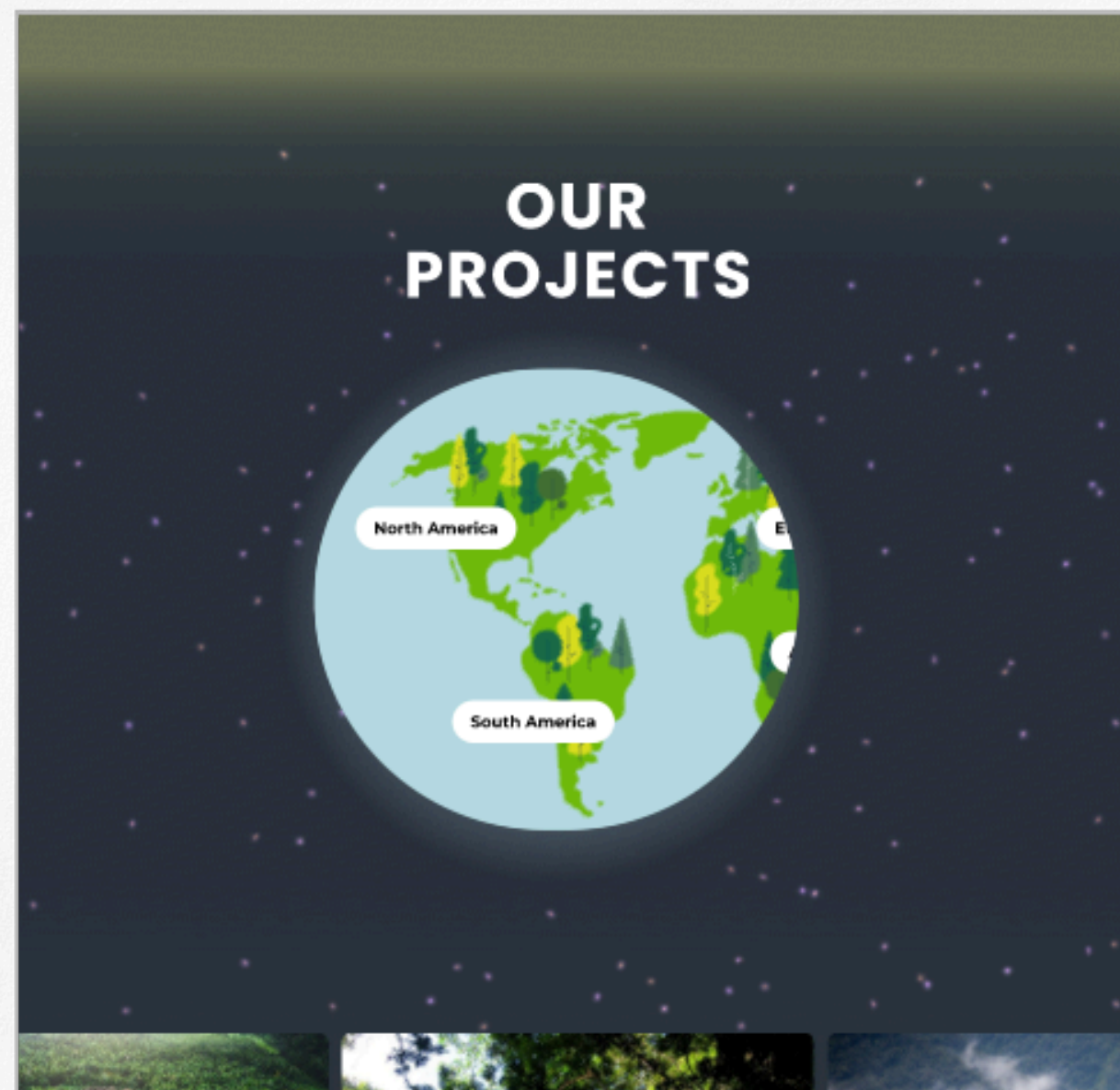


after

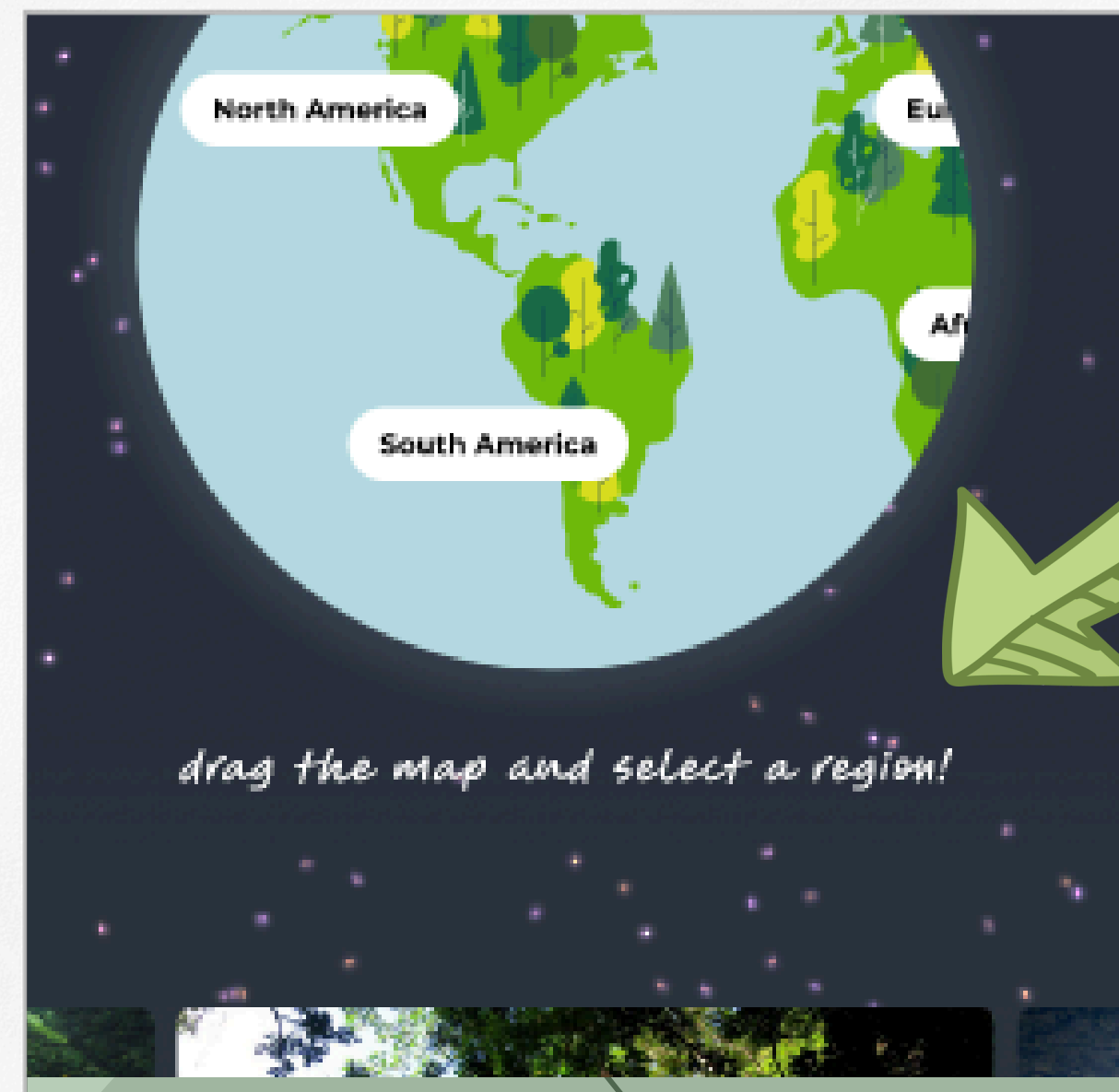
Issue Ref #10

The world map had no indication it was draggable, and users were not sure if they could scroll through.

- Users would find the map feature and be unsure what to do with it.
- The map had low **affordance** as users could not tell that the map was draggable
- Reduced the **efficiency** of the users as they could not effectively look through the map and filter through the projects via country.
- Created a **confusing** user experience.



before



after

Issue Ref #15

The contrast ratio of the text in the leaderboard toggle wasn't sufficient which minimized its visibility.

- WCAG guidelines require a 4.5 contrast ratio for small text, in which the demo had a ratio of 4.4. This is important as it increases **readability** and promotes accessibility to improve user experience.
- Reduces the **affordance** of the buttons as the reduced contrast made it harder for users to see and perceive the button as clickable.
- Affects the **effectiveness** of the button as it is less likely to be clicked to reduced affordance, as well as the **efficiency** as users may take longer to realise the button has clicking functionality.
- Creates an **annoying** user experience as well as making the site **less aesthetically pleasing**.

ATION LEADERBOA

SEARCH



MOST RECENT

MOST TREES

Anonymous

20 trees

7/22/2024, 10:16:19 AM

Anonymous

2 trees

7/22/2024, 7:06:31 AM

Mrniceguy142

Keep on going.

100 trees

7/22/2024, 6:47:38 AM

HANGFLOW FAMILY

before

ATION LEADERBOARD

SEARCH



MOST RECENT

MOST TREES

Anonymous

20 trees

7/22/2024, 10:16:19 AM

Anonymous

2 trees

7/22/2024, 7:06:31 AM

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Keep on going.

100 trees

7/22/2024, 6:47:38 AM

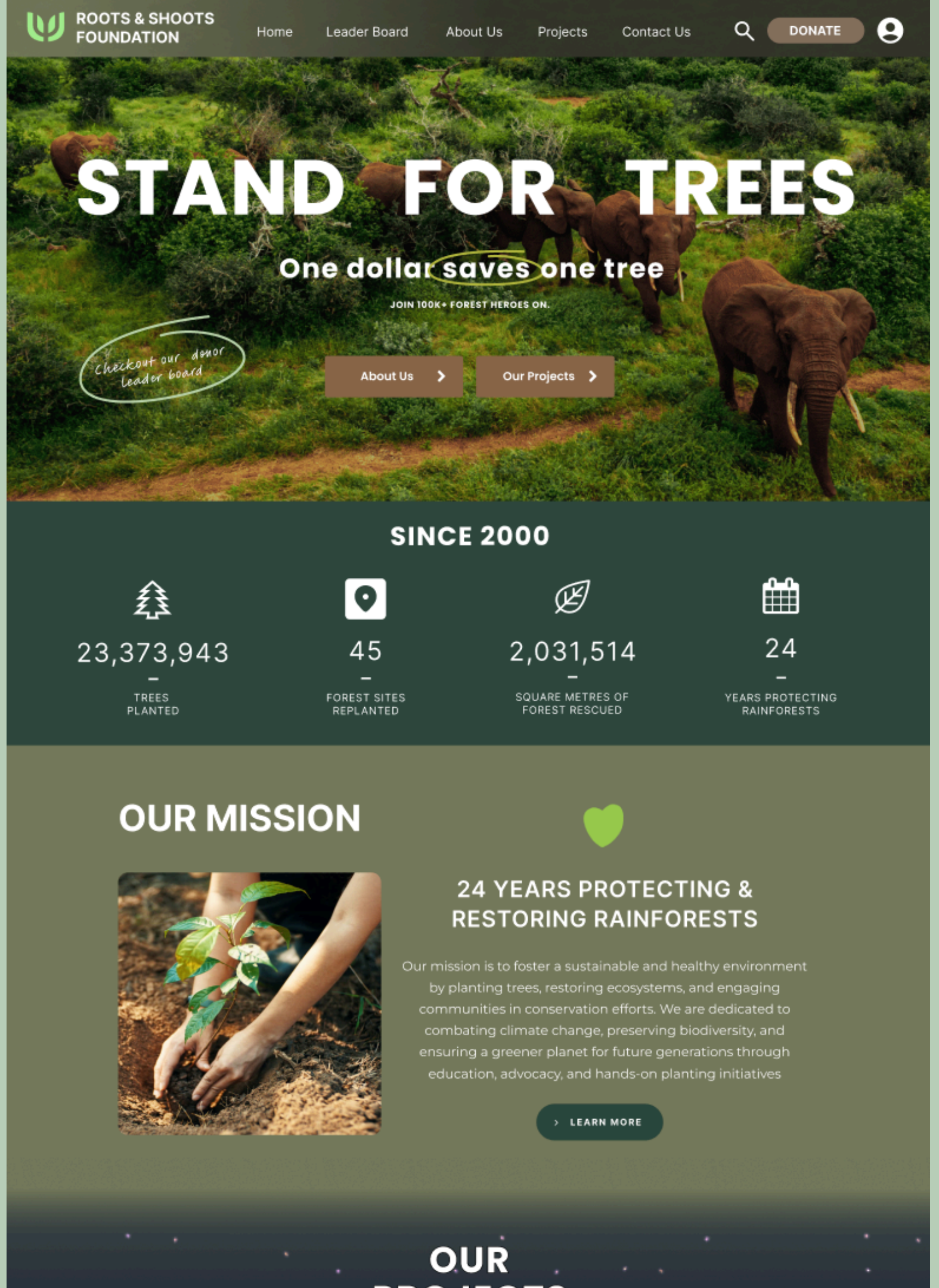
HANGFLOW FAMILY

HangFlow ry

5 trees

after

Revised Prototype



13,000 square km of the Amazon Rainforest was **deforested** in 2022

17 - 20% reduction in the forestry of the Rainforest since the 1970's

8% of the Rainforest has been **destroyed** in the past 20 years



 show transcript sign language interpreter ☐

 Search Transcript

00:00 [Narrator] Forests cover about 30% of the planet

00:04 And the ecosystems they create play an essential role

00:07 in supporting life on earth.

00:09 But deforestation is clearing earth's forest

00:12 on a massive scale.

00:14 And at the current rate of destruction,

00:16 the world's rainforest can completely disappear

OUR GOAL:

150,000

TREES

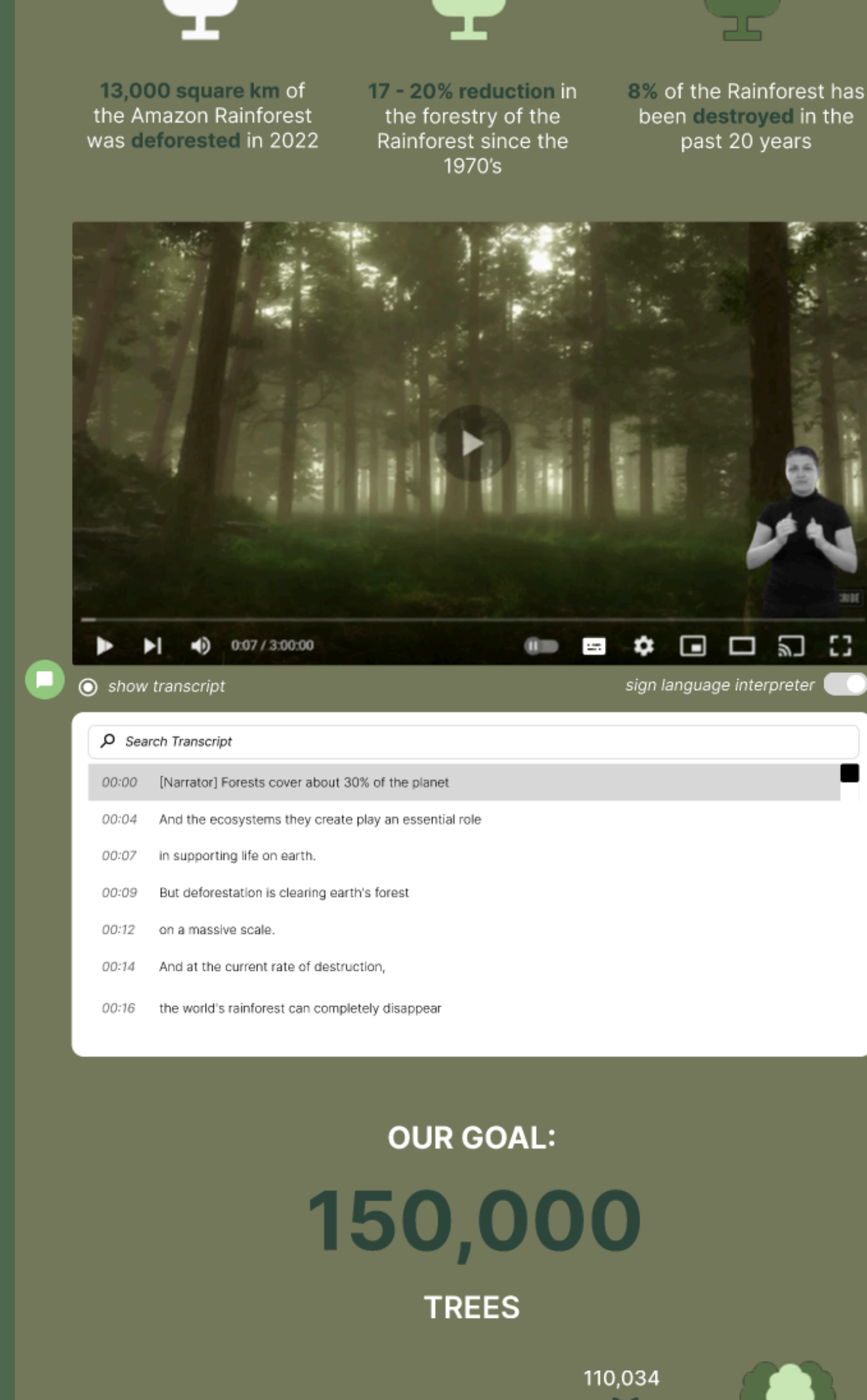
110,034



Accessibility Features

Video Transcript

all audio elements on the site contain an option to Show Transcripts, allowing hearing-impaired users to engage with content through sight rather than sound



Sign Language Interpreter

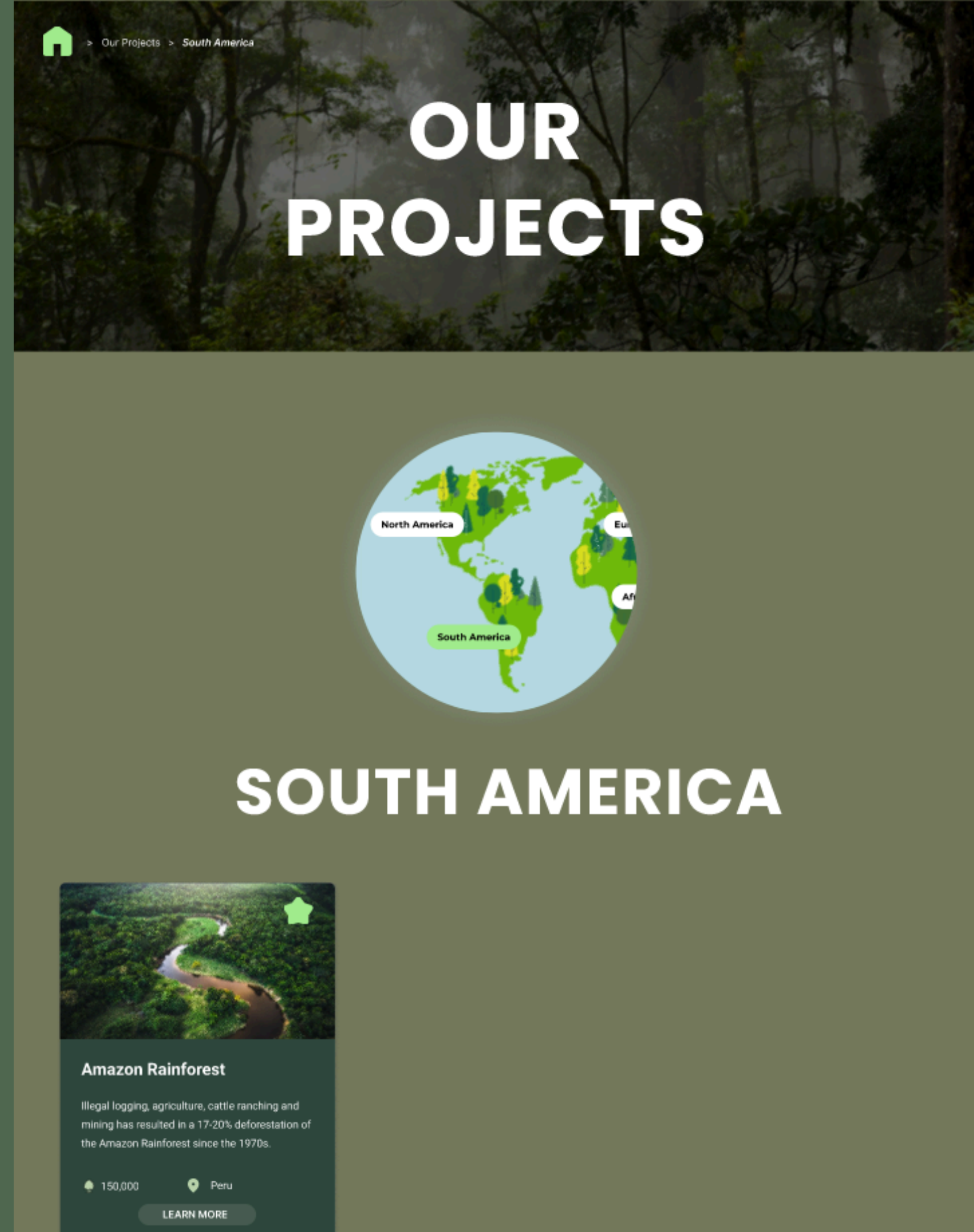
all video content on the site contains an option to show a pre-recorded sign language interpreter on screen, providing impaired users with another method of following content

Video Subtitle Options

all video content on the site contains an subtitle option that translates audio content into text on screen within the video

Visual Cues

bold and distinct colours are used to emphasise user actions (region button selection, favouriting Projects, Home button) to provide visual cues of selections



OUR GOAL:

150,000

TREES

110,034



CO₂

increasing forest cover will promote the Amazon's role as a carbon sink and offset CO2 emissions



replanting will replenish degraded areas of the Rainforest, which can restore biodiversity and stabilize ecosystems



more trees will positively influence local precipitation patterns and maintain humidity levels globally and regionally



planting native tree species can provide a habitat for a wide range of animal and plant species

BACK TO TOP ↑



ROOTS & SHOOTS
FOUNDATION

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[Leader Board](#)
[About Us](#)
[Projects](#)
[Contact Us](#)



Subscribe for updates

First Name

JAMES

Last Name

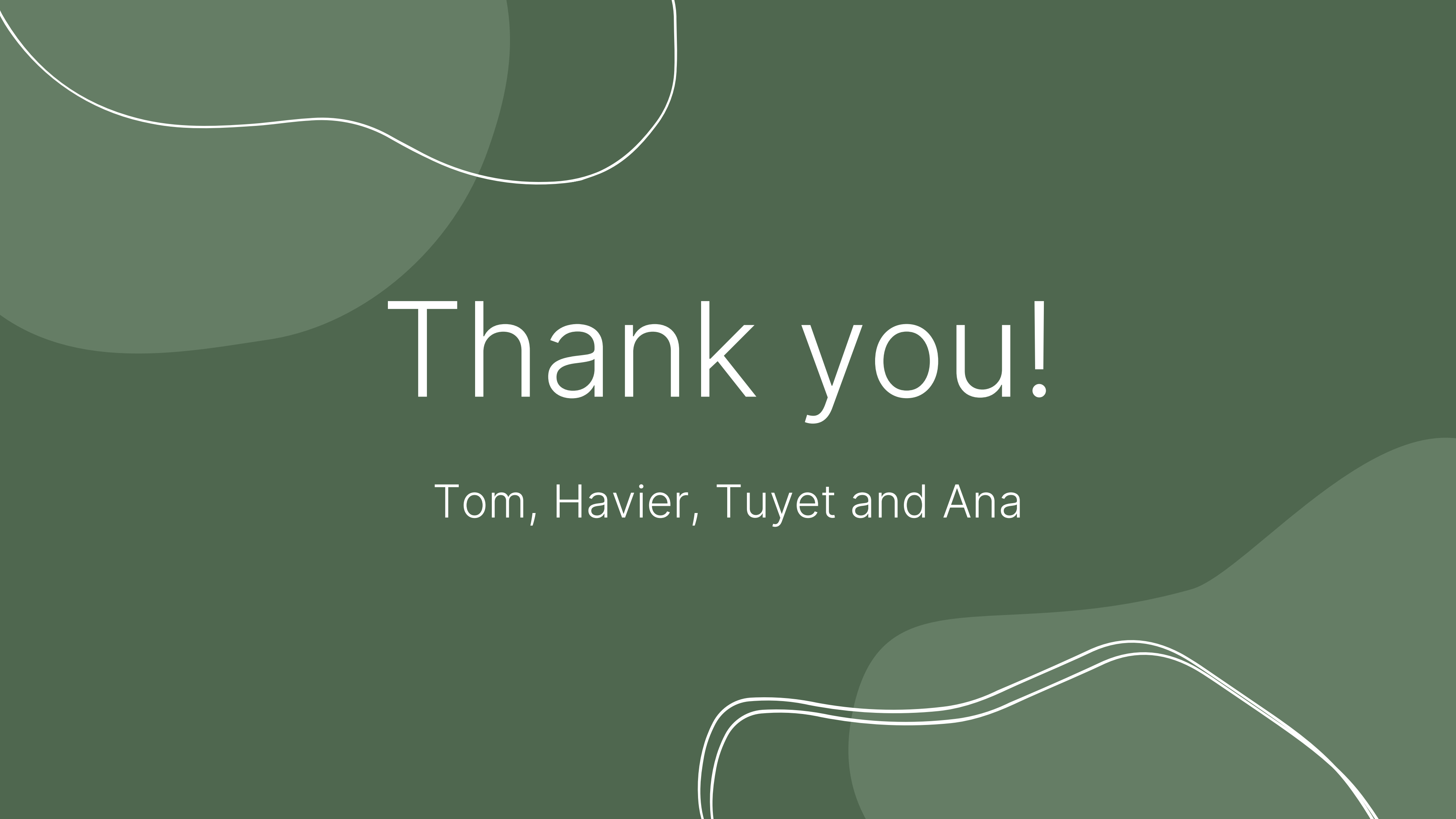
DOE

Email

JAMES.DOE@EXAMPLE.COM

Visual Icons

icons and images are used comprehensively throughout the site to assist in associating textual information into an easier-to-process visual form

The background is a solid dark green color. It features several abstract white wavy lines that flow across the frame. One line starts in the top left and curves towards the center. Another line starts in the bottom right and curves towards the center. There are also some lighter green, semi-transparent circular or oval shapes layered behind the main green background.

Thank you!

Tom, Havier, Tuyet and Ana